

PROVING TEST ON THE RELIABILITY FOR NUCLEAR FUEL ASSEMBLIES

A translation of excerpt

Summary Report of
Proving Tests on the Reliability
for Nuclear Power Plant - 1989

Nuclear Power Engineering Test Center

This report is a translation of excerpt from "The Summary Report of Proving Tests on the Reliability for Nuclear Power Plant-1989 (Japanese version published by Nuclear Power Engineering Test Center and Japan Power Engineering and Inspection Corporation).

PROVING TEST ON THE RELIABILITY FOR NUCLEAR FUEL ASSEMBLIES

This proving test is conducted by NUPEC under the sponsorship of the Ministry of International Trade and Industry (MITI), in order to verify the reliability of fuel assemblies used for commercial nuclear power plants. This test comprises three parts: the fuel irradiation test, the maximum thermal loading test and the void fraction measurement test in fuel rod bundles.

1) Irradiation Test

The purpose of the irradiation test is to demonstrate the reliability of BWR and PWR fuels in Japan. BWR fuel test was carried out in fiscal year 1976-1985 and PWR fuel test was carried out in fiscal year 1977-1986.

The monitoring fuel assemblies were fabricated by normal process, on which extra examinations were carried out to collect the pre-irradiation data. These fuel assemblies were irradiated under normal operational conditions, for which operational data were obtained. After every irradiation cycle, the reactor site inspection was carried out, and one or two assemblies were selected and submitted to the detailed post-irradiation examination at a hot laboratory.

On the basis of these data, fuel behaviors under irradiation were evaluated, and thus the safety and reliability of fuel assemblies were verified.

Please refer to the separate report for the

details of the irradiation test.

2) Maximum Thermal Loading Test

The maximum thermal loading test program for BWR and PWR standard fuel assemblies were started in 1978 to prove the reliability of the fuel and thermal hydraulic design method.

For this test on BWR fuel assemblies, the out-of-pile test facility was installed at the Isogo Engineering Laboratory of the NUPEC. Tests using simulated fuel assemblies started since the beginning of 1985, and completed by evaluating the test results in fiscal year 1986.

As of the PWR, the out-of-pile test facility was installed at the Takasago Engineering Laboratory of NUPEC. The test completed by evaluating the test results in fiscal year 1989.

3) Void Fraction Measurement Test in Fuel Rod Bundles

In order to prove the adequacy of fuel and thermal hydraulic design and the safety of nuclear power reactors, the void fraction (ratio of steam to water) in fuel rod bundles simulating the heat generation in a reactor is measured, using thermal hydraulic test facilities which can simulate the high-temperature and high-pressure conditions in a reactor. Tests are carried out for both BWR and PWR fuel assemblies. The basic design of the tests was performed and the manufacture of the test facility for BWR fuel was started in 1987.

I. MAXIMUM THERMAL LOADING TEST

The maximum thermal loading test is intended to prove the reliability of fuel assembly design by confirming the thermal margin of heat transfer.

In this test, one test facility, which can simulate the high temperature and high pressure conditions similar to the power reactor is built for each BWR and PWR fuel assembly. The heat generation of the fuel assembly in the power reactor is simulated by the electric heating of clad instead of nuclear fission.

In light water reactors, electricity is generated by the turbine generator, rotated by the steam generated by heating the coolant with the heat produced by nuclear fission of the fuel assembly. The amount of the heat which is transferred from the fuel assembly to the coolant depends on the flow rate and the temperature of

the coolant, the surface temperature of the clad, and the amount of bubbles in the coolant.

In order to safely transfer the heat from the fuel assembly, the amount of the heat must be less than a thermal limited value (a maximum thermal load). If it exceeds the limit, the fuel assembly might get burnt out. In order to keep the fuel assembly integrity, the removal of the heat from the fuel assembly must be performed safely not only in the normal operation of the power reactors but also in the abnormal transient.

This proving test, using a simulated fuel assembly is carried out to verify the tolerance in the removal of the heat from the fuel assembly and thus prove the reliability of fuel and thermal hydraulic design.

1. BWR Fuel Assembly Maximum Thermal Loading Test

[1] Outline of the Test

This test is performed to prove the reliability of the fuel and thermal hydraulic design method, by measuring the critical power of BWR fuel assemblies under steady and transient conditions, the coolant flow characteristics and the void characteristics, with parametrically used the coolant conditions and the shape, etc. of fuel assemblies.

This test was made by the test facility which could simulate the high temperature and high pressure conditions of the power reactor using 8×8 mock-up fuel assemblies which could simulate the heat generation of fuels in the power reactor by electric heating instead of nuclear fission. Table 1 · 1 shows the principal features of the test facility installed at the Isogo Engineering Laboratory.

This project began with the fundamental design of the test facility in 1978. Construction was completed on December, 1984. The proving test started from January, 1985 and the test and evaluation were completed on March, 1987.

[2] Contents of the Test

The following four tests were carried out.

(a) Maximum thermal loading test under steady conditions

The critical power of the full scale mock-up fuel assemblies was measured, using the key parameters of fuel shape, axial and radial power distributions, pressure, flow rate and inlet subcooling. The three parameters pressure, flow rate and inlet subcooling were varied not only within the actual reactor operation range but also beyond the actual reactor operating range and the transient analysis evaluation limit.

(b) Maximum thermal loading test under transient conditions

The critical power of the full scale mock-up fuel assemblies was measured by simulating the changes in

the power, pressure, coolant flow rate and inlet enthalpy under the typical thermally-severe transient conditions supposed in an actual reactor.

Transient conditions tested were as follow.

- Loss of feed water heating.
- Recirculation pump trip.
- Turbine trip without bypass.

(c) Flow characteristic test

Pressure loss within the fuel assemblies, in both single phase flow and two phase (steam-water) flow, was measured under the thermal hydraulic conditions, including the actual reactor operating range.

(d) Void characteristic test

Under the thermal hydraulic conditions covering the actual reactor operating range, the coolant void fraction was measured by an X-ray densitometer at the outlet piping of the test section loading the full scale mock-up fuel assemblies.

The estimation procedure flow for each test item is shown in Figure 1 · 1.

[3] Specification of the Test Assembly

The test assemblies for the tests (simulated fuel assemblies) simulate a new 8×8 and a new 8×8 retrofit fuel configuration on a full scale. The 62 simulated fuel rods, distributed in a tetragonal array, are directly heated by direct current. Thermocouples are installed in the simulated fuel rods to detect the Boiling Transition (B. T.).

The axial power distribution of the simulated fuel rods is assumed to be a chopped cosine distribution in most cases, which is a typical distribution of BWR. And, some tests with an inlet peak distribution are also performed to examine the critical power and void char-

Table 1 · 1 Principal features of test facility for BWR fuel assembly

Item	Specification
Maximum power of test section	12 MWt (about 200% of the hottest bundle of BWR)
Maximum flow rate	75 t/h (about 130% of the bundle flow of BWR)
Maximum pressure	95 kg/cm ² a (about 130% of rated pressure of BWR)

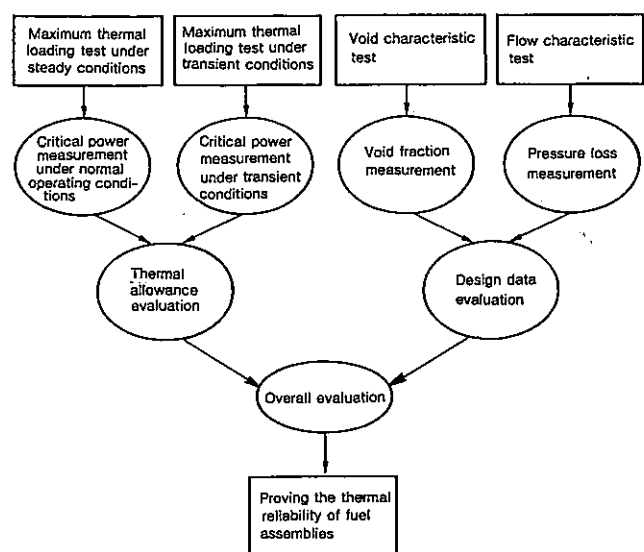


Fig. 1 · 1 Estimation procedure in maximum thermal loading test on BWR fuel assemblies

Table 1・2 Principal specifications for BWR test assembly

Item		Specification	
Type of test assembly		New 8×8 Retrofit	New 8×8 type
Size of array		Full array	Full array
Number of heater rods		62	62
Heater rod outer diameter	[mm]	12.3	12.3
Active length	[m]	3.708	3.708
Number of water rods		2	2
Water rod outer diameter	[mm]	15.0	15.0
Pitch of heater rod	[mm]	16.3	16.2
Channel box inner size	[mm]	134.1	132.5

Table 1・3 Items of test assembly for BWR

Test assembly No.		I	II, III	IV	V
Lattice type		New 8×8 retrofit	New 8×8	New 8×8	New 8×8
Power distribution	Axial	Cosine	Cosine	Inlet peak	Cosine
	Planar	"A"*	"A"	"A"	"B"***
Test item	Steady state critical power	142 data	256 data	97 data	97 data
	Transient critical power		60 cases	32 cases	
	Void fraction		87 data	32 data	
	Two-phase flow characteristics	35 data	35 data		
	Single-phase flow characteristics	24 data	24 data		
Total		201 data	462 data	132 data	97 data

*"A": Simulation pattern for beginning of operation

***"B": Simulation pattern for middle of operation

acteristics dependency on the axial power distribution. The simulated fuel rods have a constant diameter with tubes having various thickness values which provide for distribution of the electric resistance along the axial direction to simulate the axial power distribution.

Two radial power distributions of the test fuel assembly are simulated. One (distribution A) is simulated the beginning of life of fuel in an example of the BWR fuel design, which has the largest local peaking factor. The other (distribution B) is simulated the middle of life of the fuel.

Geometry specifications for the BWR fuel assembly are shown in **Table 1・2**, and the test assembly items are shown in **Table 1・3**.

[4] Outline of Test Facility

The test facility consists of three systems : the high-temperature and high-pressure loop system, the electric power supply system, and the instrument and control system.

A bird's-eye view for the test facility is shown schematically in **Figure 1・2**.

(a) The high-temperature and high-pressure loop system

The system diagram of the facility is shown in

Figure 1・3. The specifications for main test facility components are shown in **Table 1・4**.

1) Test section The cross-section of the test section is shown in **Figure 1・4**. The test section consists of a pressure vessel, a ceramic simulated channel, and electrodes set at an upper and lower part. It can be equipped with a BWR simulated fuel assembly of the same size as of the real reactor, to assimilate the same thermal hydraulic conditions as in a BWR. In addition to measure the heater tube temperature for the fuel rods, measurement of the pressure drop for the cooling water is facilitated by pressure tap tube. The outlet piping of the test section has three holes for void fraction measurement.

2) Steam drum Pressure control of the system is done by a steam drum. The circulating coolant, which becomes a two phase flow of steam and water in the test section, is separated into steam and saturated water by a gas-liquid separator in the steam drum. The steam is condensed by the spray water from the upper nozzle of the drum to keep the system pressure constant. System pressure in both steady and transient tests is controlled by spray lines, which have four different sized valves.

3) Air-cooled heat exchanger The heat gener-

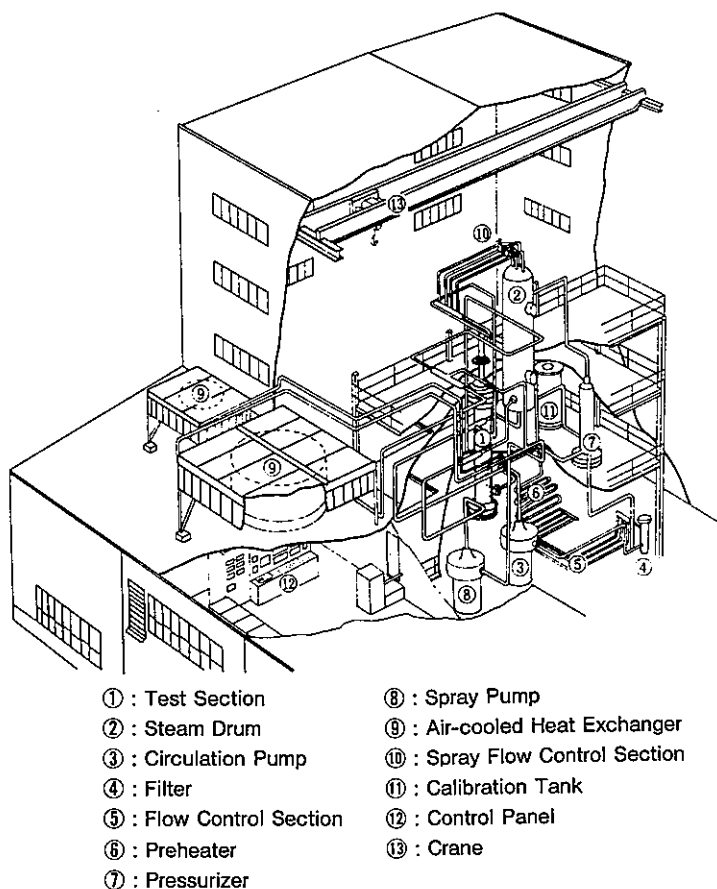


Fig. 1-2 Bird's eye view of the BWR fuel assembly maximum thermal loading test facility

ated in the system is removed to the air by two air-cooled heat exchangers. Water, cooled by the air-cooled heat exchangers, is used as spray water in the steam drum, as described above and condenses the steam.

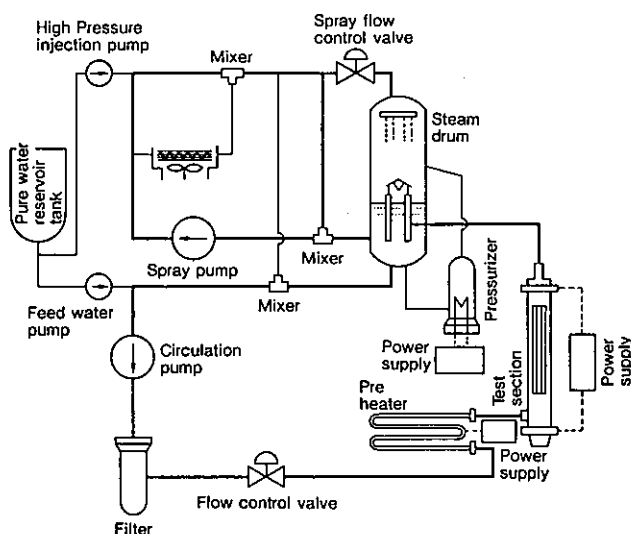


Fig. 1-3 System diagram for BWR maximum thermal loading test facility

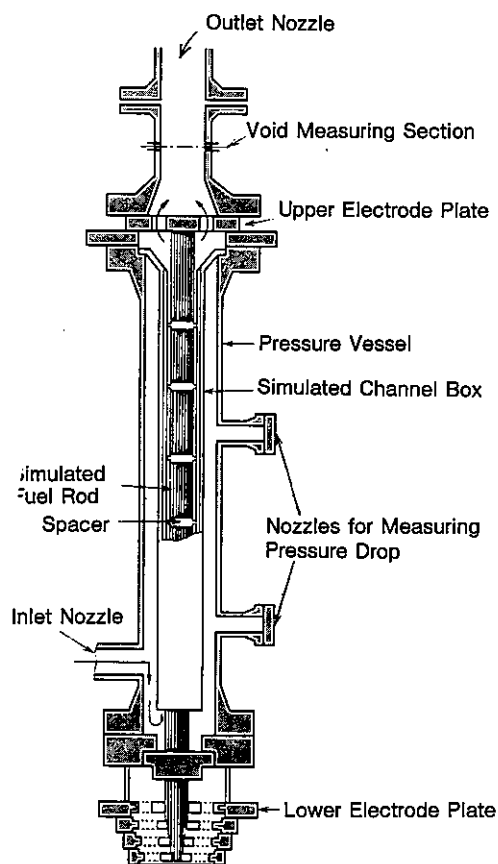


Fig. 1-4 Cross-sectional view showing of test section of BWR fuel assembly maximum thermal loading test facility

4) Pre-heater In order to control the inlet temperature of the coolant in the test section, which is one of the test parameters, a pre-heater is equipped. The pre-heater controls the inlet sub-cooling with heated piping by direct current.

5) Pressurizer A pressurizer is used for fine controlling of the pressure, when test assembly power is low.

6) Pumps Main pumps are the circulation pump to supply cooling water to the test section, and the spray pump to supply spray water to the steam drum. Others are auxiliary components cooling water pump, high-pressure water pump and test section electrode cooling water pump.

7) Other components These components include the circulation system filter, weighing tank, silencer, mixer to mix the water having different temperatures, compressor to supply compressed air, and demineralizer to supply the demineralized water.

(b) Electric power supply system

The electric power supply system consists for the test section, for the pre-heater, for the pressurizer, and for the other units. Power supplies, which are used as heat sources, such as for the test section, are operated

Table 1 - 4 Specifications for main components in BWR fuel assembly maximum thermal loading test facility

	Components	Specifications		Components	Specifications
Components	Test section	Type : Vertical cylindrical type Design pressure : 105 kg/cm ² a Design temperature : 314°C Incorporating mock-up ceramic channels	Power source	Pre-heater thyristor	Type : Indoor, sealed type Rated DC output : 630 kW Rated DC voltage : 45 V Rated DC current : 14000 A
	Steam drum	Type : Vertical cylindrical type Design pressure : 105 kg/cm ² a Design temperature : 314°C		Pressurizer thyristor	Type : Indoor, open type Rated DC output : 300 kW Rated DC voltage : 400 V Rated DC current : 750 A
	Air-cooled heat exchangers (A & B)	Type : Lateral, forced ventilation type Design pressure : 105 kg/cm ² a Design temperature : 314°C Maximum heat exchange : A 9.8 MW B 3.4 MW	Instrumentation	Central control panel	Type : Bench board type Approximate size : 2300 H X 3300 W X 2580 D
	Pre-heater	Type : DC direct electrification & heating type Design pressure : 125 kg/cm ² a Design temperature : 375°C Output : 600 kW		Burn-out monitoring panel	Type : Independent type Approximate size : 2300 H X 1300 W X 1000 D
	Pressurizer	Type : DC cartridge heater incorporating type Design pressure : 105 kg/cm ² a Design temperature : 314°C Output : 300 kW		Central ACU	Type : TOSBAC-7/60
	Circulation pump	Type : Canned motor type Design pressure : 125 kg/cm ² a Design temperature : 314°C Flow rate : 85 t/h Lift : 180 m		Main memory	Capacity : 512 k byte
	Spray pump	Type : Canned motor type Design pressure : 125 kg/cm ² a Design temperature : 314°C Flow rate : 100 t/h Lift : 180 m		Auxiliary memory	Fixed head disc (4 M byte) Fixed disc (29.76 M byte) Floppy disc (237.5 k byte)
				Peripheral units	System typewriter Color CRT display Line printer Magnetic tape (9 track 1600 BPI) Data recording time (3 minutes per case) Trend recorders (2 units)
Power source	Test section thyristor (2 units)	Type : Indoor, sealed type Rated DC output : 6000 kW (per unit) Rated DC voltage : 150 V Rated DC current : 40000 A	Computer system	Process I/O devices	Analog input device Superhigh-speed analog input device Contact input device Interruption input device Analog output device Relay output device Digital input device Data transfer device

from D.C. power, since fine control performance is necessary.

(c) Instrument and control system

The outline of the instrument and control data flow in this facility is shown in **Figure 1 - 5**.

1) Central control panel The equipments operation, the test parameters set-up, and monitoring the system are carried out at the central control panel. The central control panel has graphic panels, recorders,

indicators, alarms, and switches for the equipment.

2) Monitoring panel for a burn-out This panel monitors and records the temperature of the heater tube at the test assembly. The monitoring panel for a burn-out is equipped with a burn-out detectors and recorders.

3) Computer system The computer system for this facility has functions for accomplishing an automatic control at start-up/shut-down, a system control at the transient simulation test and monitoring the

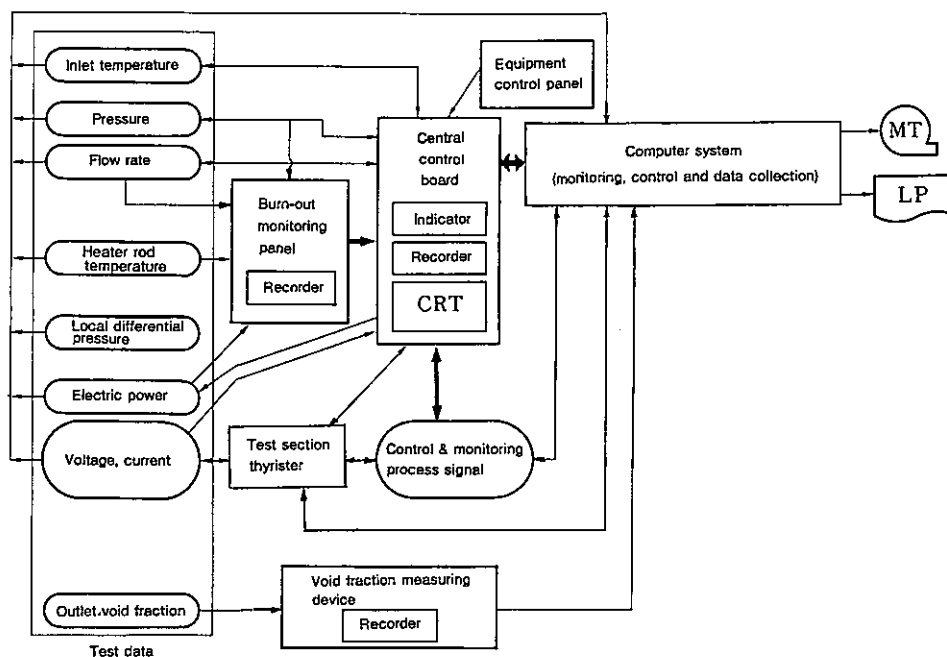


Fig. 1-5 Outline showing instrumentation control data flow in BWR fuel assembly maximum thermal loading test facility

system as well as collecting test data. Further, this system has a function for monitoring the thermal electrode signals. When the heater rods reach the set temperature, electric power is automatically reduced by about 20% to prevent damage to the heater rods.

[5] Proving Test

(a) Maximum thermal loading test under steady conditions

This test was carried out with test assemblies (I) through (V).

1) Test method Tests were conducted by setting the bundle flow rate, the inlet fluid temperature and pressure. After the desired steady state conditions sufficiently stabilized, the test was started by increasing the test assembly power in small increments. Thermocouples were used to monitor the change of the heater rod surface temperature during the test. The test assembly power was increased until the heater rod surface temperature reached the pre-set value (14°C from the stable level). The critical power was defined as the power causing the heater rod surface temperature to increase 14°C from the stable level. Further, the heater rod surface temperature was raised to about 500°C , using the test assembly (V) to evaluate the critical power sensitivity to triggering level (14°C) of the heater rod surface temperature.

2) Test conditions The test was carried out under the following conditions.

Test assembly : New 8×8 retrofit fuel assembly.

New 8×8 fuel assembly.

Axial power distribution : Cosine distribution & inlet peak distribution.

Radial power distribution : Distribution A, simulating power distribution in early fuel burnup stage.

Distribution B, simulating power distribution in the middle burnup stage.

Pressure : $40 \sim 95 \text{ kg/cm}^2\text{a}$

Flow rate : $5 \sim 75 \text{ t/h}$

Inlet subcooling : $3 \sim 30 \text{ kcal/kg}$

A total of 592 bits of critical power data were taken under these test conditions. Tables 1-5 through 1-7 show the test conditions for test assemblies (I) through (V).

3) Test results Figure 1-6 through 1-10 show comparisons between the calculated value by GEXL correlation and the measured critical power.

The comparison shows a good agreement between the test data and the prediction. Average ECPR (ECPR = (critical power by GEXL) / (measured critical power)) is almost 1.0. So, the GEXL correlation is judged capable of estimating the critical power with accuracy, irrespective of different test assembly shapes and power distributions.

Figure 1-11 shows the result of comparing the calculated critical power value by GEXL correlation with the measured value, synthesizing the test assemblies (I) through (V) data. The test values and the predicted values agreed well with each other (the average ECPR = 1.001) with standard deviation being 3.4%. It was less than the standard deviation of 3.6% considered in applying GEXL correlation to the present reactor

Table 1・5 Condition of maximum thermal loading test under steady condition for test assembly (I) (II)

Inlet subcooling [kcal/kg]		Flow [t/h]							
Pressure [kg/cm ² a]		3	7	10	12	15	20	25	30
73	5	○	○	○	◎	○	○	○	○
	10	○	○	○	○	○	○	○	○
	20	○	◎	○	◎	○	◎	○	◎
	30	○	○	○	○	○	○	○	○
	45	○	○	○	○	○	○	○	○
	55	◎	◎	◎	◎	◎	◎	◎	◎
	65	○	○	○	○	○	○	○	○
	75	○	○	○	○	○	○	○	○
56	20		○		◎		○		○
	45		○		◎		○		○
	55		◎		◎		◎		◎
	65		○		◎		○		○
88	20		○		◎		○		○
	45		○		◎		○		○
	55		◎		◎		◎		◎
	65		○		◎		○		○
40	20				○		○		
	45				○		○		
	55				◎		○		
	65				○		○		
95	20				○		○		
	45				○		○		
	55				◎		○		
	65				○		○		

◎ : Reproducibility test

core design, and verified the adequacy of the current power reactor thermal hydraulic design method.

Figure 1・12(a) shows typical heater rod temperature behaviors obtained in the test, where the heater rod temperature was raised to 500°C using the test assembly (V). **Figure 1・12(b)** shows how the boiling transition judging temperature affects the critical power evaluation, taking the temperature at the abscissa and the over power ratio* at the ordinate. In the figure, point ● represents the standard judging temperature 14°C at which the over power ratio is 1.0. The figure indicates that the critical power sensitivity to the boiling transition judging temperature and the selection of 14°C is reasonable.

Next, **Figure 1・13** shows a comparison between the calculated value by the typical correlation CISE and the test results. The comparison shows a worse agreement than that with the GEXL correlation in **Figure 1・**

(Note) *Over power ratio : When the boiling transition judging temperature is changed, the term shows the ratio of the critical power judged at that temperature to the critical power judged at the standard boiling transition judging temperature (14°C).

Table 1・6 Condition for maximum thermal loading test under steady condition for test assembly (III)

Inlet subcooling [kcal/kg]		Flow [t/h]							
Pressure [kg/cm ² a]		3	7	10	12	15	20	25	30
73	5				○				
	10	◎	◎	◎	◎	◎	◎	◎	◎
	20	○	○	○	◎	○	○	○	○
	30	◎	◎	◎	◎	◎	◎	◎	◎
	45	○	○	○	◎	○	○	○	○
	55	◎	◎	◎	◎	◎	◎	◎	◎
	60	○	○	○	○	○	○	○	○
88	30	○	○	○	○	○	○	○	○
	55	○	○	○	◎	○	○	○	○
	60	○	○	○	○	○	○	○	○

◎ : Reproducibility test

Table 1・7 Condition for maximum thermal loading test under steady condition for test assembly (IV) (V)

Inlet subcooling [kcal/kg]		Flow [t/h]							
Pressure [kg/cm ² a]		3	7	10	12	15	20	25	30
73	5	○	○	○	○	○	○	○	○
	10	○	○	○	○	○	○	○	○
	20	○	○	○	○	○	○	○	○
	30	○	○	○	○	○	○	○	○
	45	○	○	○	○	○	○	○	○
	55	○	○	○	◎	○	○	○	○
	60	○	○	○	○	○	○	○	○
88	30	○	○	○	○	○	○	○	○
	55	○	○	○	○	○	○	○	○
	60	○	○	○	○	○	○	○	○

◎ : Reproducibility test

11. Comparison was also made on other correlations than CISE. These comparison indicated that the GEXL correlation had estimated the test result with higher accuracy than any other published correlations. **Figures 1・14 and 1・15** show comparisons between the calculated value by the subchannel codes and the test result. The estimation precision is worse than that with the GEXL correlation, but it is found that, if improved in future, the subchannel codes could estimate the critical

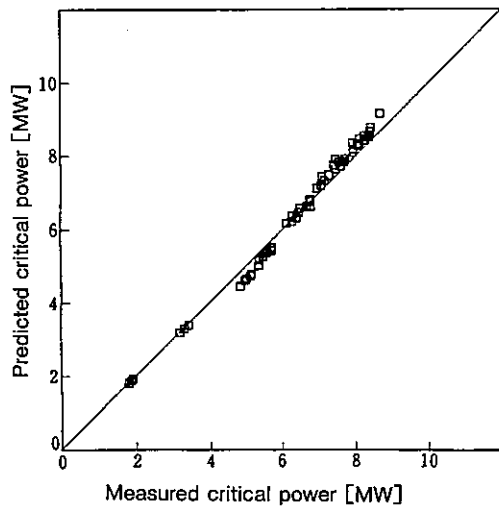


Fig. 1・6 Critical power data vs. GEXL prediction for test assembly (I) (GEXL application range)

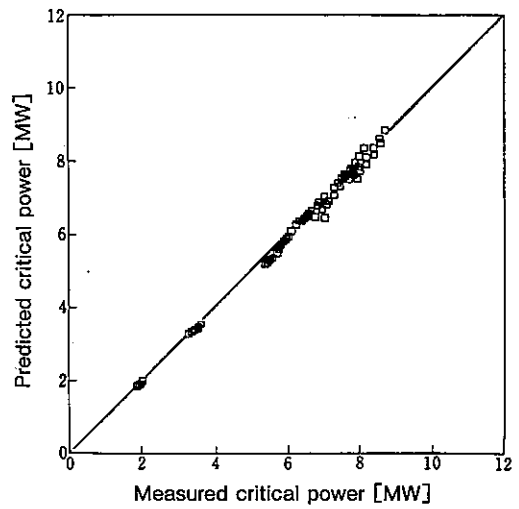


Fig. 1・9 Critical power data vs. GEXL prediction for test assembly (IV) (GEXL application range)

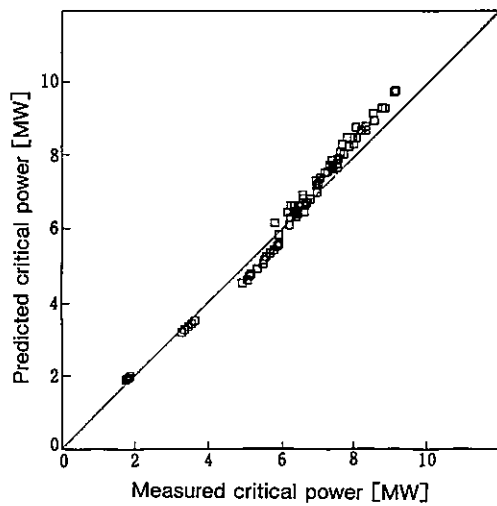


Fig. 1・7 Critical power data vs. GEXL prediction for test assembly (II) (GEXL application range)

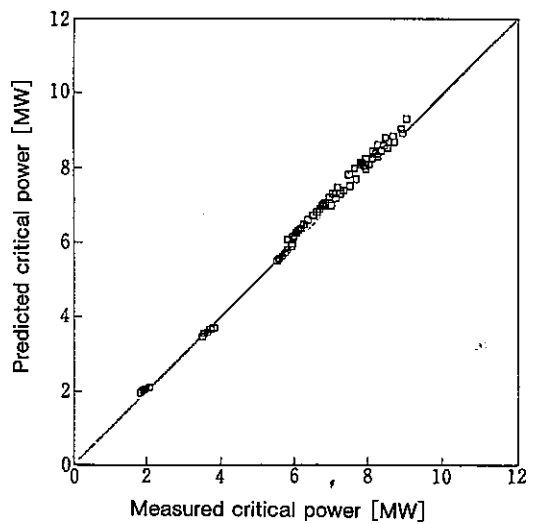


Fig. 1・10 Critical power data vs. GEXL prediction for test assembly (V) (GEXL application range)

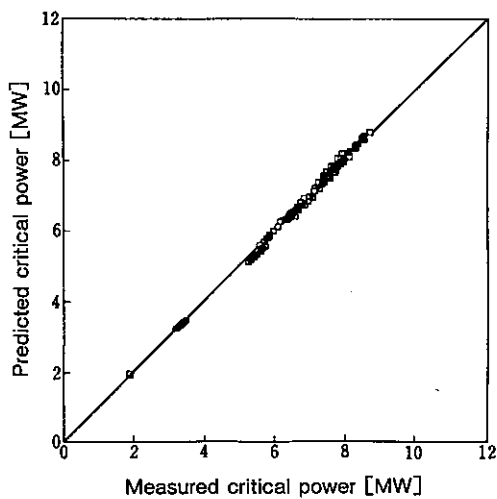


Fig. 1・8 Critical power data vs. GEXL Prediction for test assembly (III) (GEXL application range)

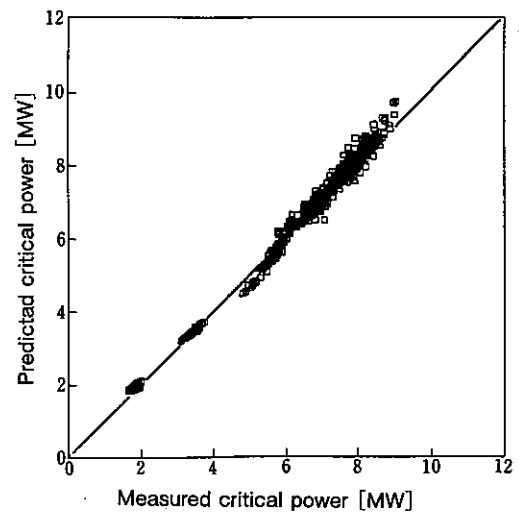


Fig. 1・11 Critical power data vs. GEXL prediction for all test assemblies (GEXL application range)

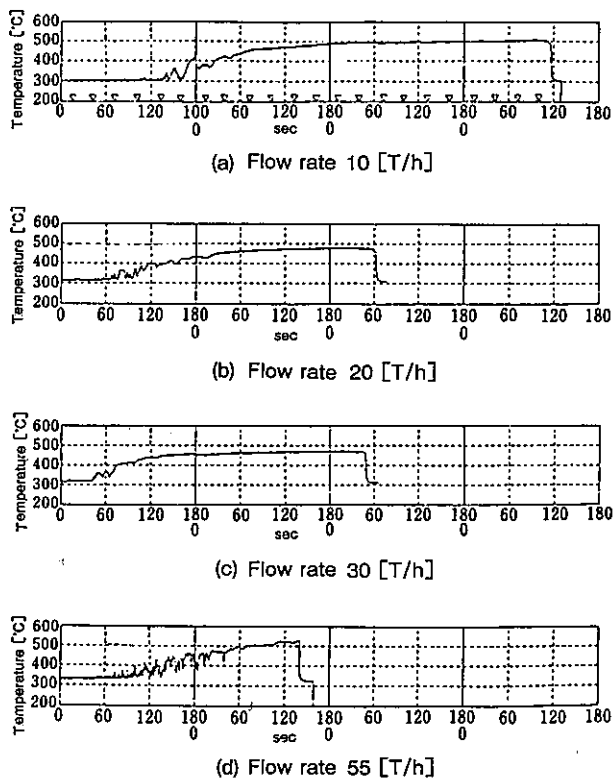


Fig. 1-12 (a) Typical heater rod temperature behaviors at boiling transition judging temperature evaluation test (test assembly (V))

power of fuel assembly with accuracy.

(b) Maximum thermal loading test under transient conditions

The loss of feed water heating, the recirculation pump trip and the turbine trip without bypass were selected as the thermally-severe transient conditions expected in an actual reactor. The critical power of fuel assemblies under transient conditions was measured, simulating changes in power, pressure, coolant flow rate and inlet enthalpy with the passage of time. The test assemblies used were (III) with the cosine axial power distribution, and (IV) with the inlet peak distribution. This test collected data mainly for boiling transition, including cases where boiling transition was not reached.

1) Test method In the test, test parameters such as the coolant inlet enthalpy, flow rate, pressure and test assembly power were directly controlled by the process computer, as expected throughout the transient, to observe whether boiling transition occurs or not.

2) Test conditions The test was performed within the following range.

Test assembly: New 8×8 fuel assembly

Axial power distribution: Cosine distribution & inlet peak distribution.

Radial power distribution: Distribution A, simulating

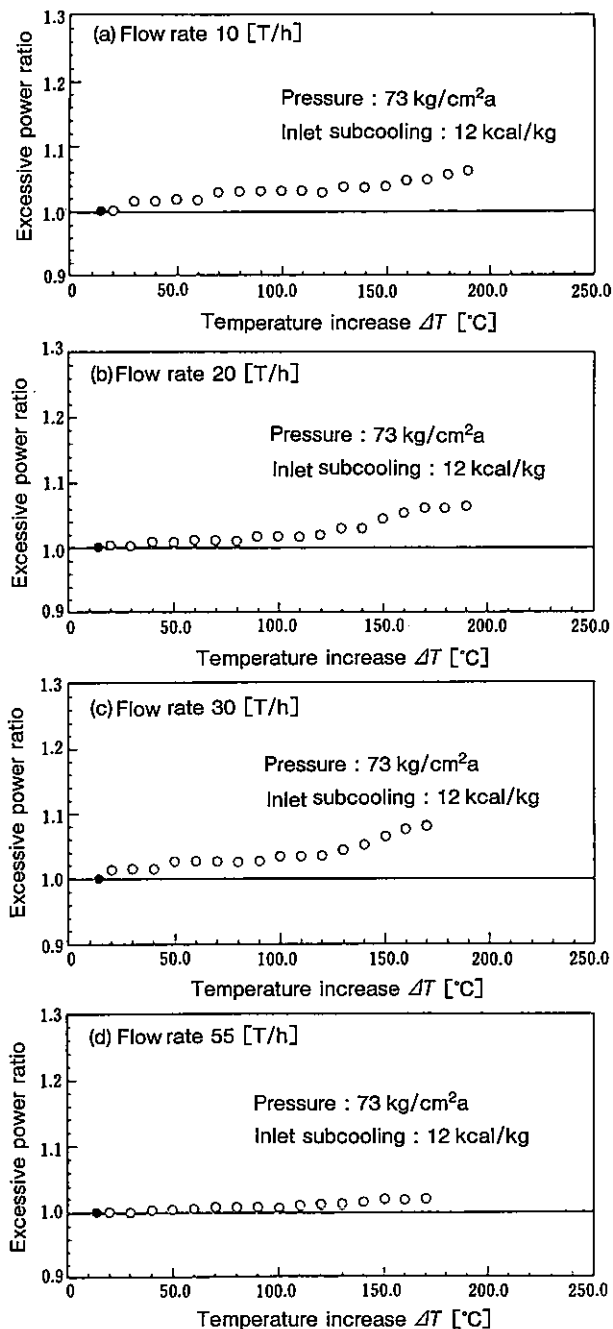


Fig. 1-12 (b) Relations between boiling transition judging temperature evaluation test result ΔT and critical power ratio (test assembly (V))

power distribution in early fuel burnup stage.

Initial pressure : 73 kg/cm²a

Initial flow rate : 47~60 t/h

Initial inlet enthalpy : 293 kcal/kg

A total of 63 transient tests were carried out within the above range, including 40 tests where the boiling transition occurred. Test conditions are shown in Tables 1-8 and 1-9.

3) Test results The parameter changes in the

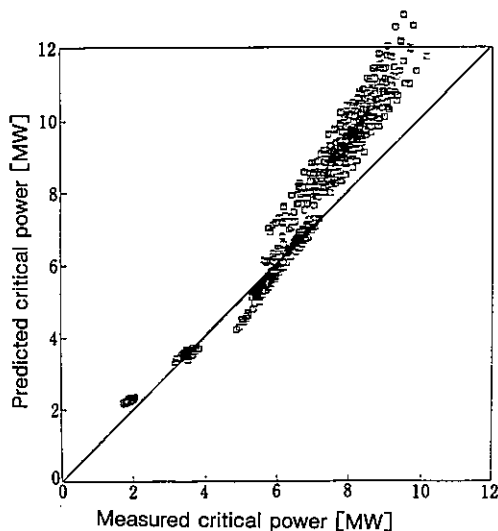


Fig. 1 · 13 Critical power data vs. CISE prediction for all test assemblies

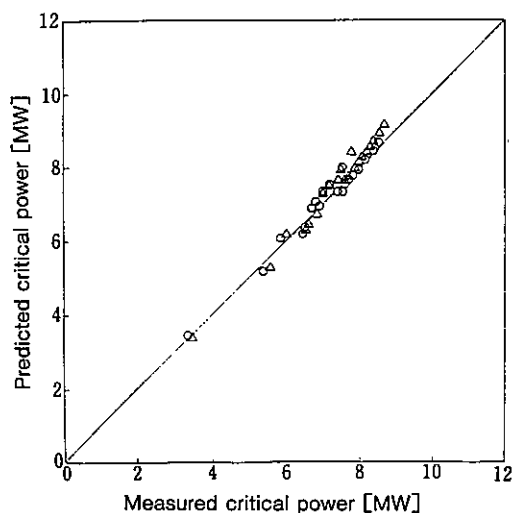


Fig. 1 · 14 Critical power data vs. Improved COBRA/BWR prediction for test assembly (III), (IV)

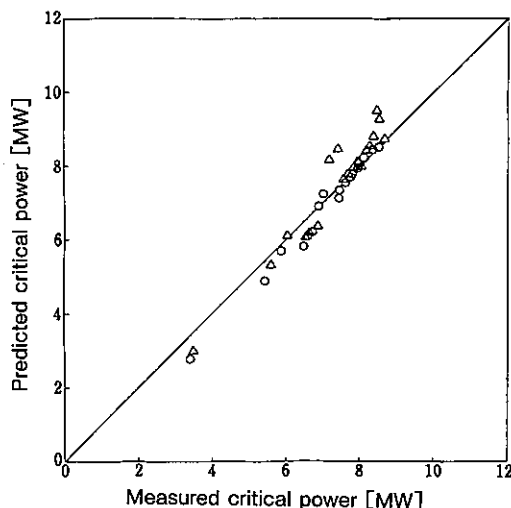


Fig. 1 · 15 Critical power data vs. Improved MULTI prediction for test assembly (III), (IV)

analysis conditions contained in the application for BWR licensing (hereinafter called the BWR fuel analysis conditions) were simulated, regarding the test assemblies whose axial power distribution was cosine distribution and inlet peak distribution, and the boiling transition was confirmed not to occur. Further, analysis results calculated by the single channel thermal hydraulic analysis code were compared with the measurement result, in respect to boiling transition onset time and ΔC_{PR} . The results are shown in Figures 1 · 16 through 1 · 20. These results show fairly good coincidence between the analysis value with measured value within the GEXL correlation precision range, and have verified the reliability of the present thermal hydraulic evaluation method for transient.

(c) Flow characteristic test

This test was carried out with test assemblies (I) and (II).

1) Test method Pressure losses were measured for single phase flow and two phase flow through the test assembly, while pressure in the test section, flow rate, inlet fluid temperature and test assembly power (zero in the single phase flow) were kept constant.

2) Test conditions Test conditions were as follow.

Test assembly: New 8×8 retrofit fuel assembly.

New 8×8 fuel assembly.

Axial power distribution: Cosine distribution.

Radial power distribution: Distribution A, simulating power distribution in early fuel burnup stage.

The two phase flow characteristic test covered:

Pressure: 73 and 88 kg/cm² a

Flow rate: 20~65 t/h

Inlet subcooling: 12 kcal/kg

Outlet quality: 0.07~0.25

Seventy bits of data were collected within the above range.

The single phase flow characteristic test covered:

Pressure: Ordinary pressure and 73 kg/cm² a

Flow rate: 20~75 t/h

Inlet subcooling: Room temperature to 12 kcal/kg
Forty eight bits of data were collected within the above range. The test conditions are shown in Tables 1 · 10 and 1 · 11.

3) Test results The measured total pressure loss at the heat generating part in the two phase flow and the single phase flow were compared with the calculated value by the design method. The results are shown in Figures 1 · 21 through 1 · 24. In the two phase flow (Figures 1 · 22 and 1 · 24), the design method predicted the measured value within a 3% relative error. In the single phase flow (Figures 1 · 21 and 1 · 23), on the other hand, the estimation precision further improved, predicting the measured value almost

Table 1・8 Conditions for transient test for test assembly (III)

Simulated transient change		Turbine trip without bypass	Recirculation pump trip	Loss of feed water heating
Basic parameter change	Pressure	Increase → decrease	Constant	Constant
	Flow rate	Decrease	Decrease	Constant
	Inlet enthalpy	Constant	Constant	Decrease
	Power	Increase → decrease	Decrease	Increase
Initial value	Pressure	73 kg/cm ² a	73 kg/cm ² a	73 kg/cm ² a
	Flow rate	3 data (85~110%)	3 data (85~110%)	3 data (85~110%)
	Inlet enthalpy	abt. 293 kcal/kg	abt. 293 kcal/kg	abt. 293 kcal/kg
	Power	≥5~6 MW	≥5~6 MW	≥5~6 MW
Change range	Pressure	Constant, 1.0 → 1.10	Constant	Constant
	Flow rate	Constant, 1.0 → 0.6	1.0 → 0.3, 0.6	Constant
	Inlet enthalpy	Constant	Constant	1.0 → 0.96
	Power	1.0~1.15	Constant, 1.0 → 0.6	1.0 → 1.15, 1.2
Sub-total		20 data	20 data	20 data

Table 1・9 Conditions for transient test for test assembly (IV)

Simulated transient change		Turbine trip without bypass	Recirculation pump trip	Loss of feed water heating
Basic parameter change	Pressure	Increase → decrease	Constant	Constant
	Flow rate	Decrease	Decrease	Constant
	Inlet enthalpy	Constant	Constant	Decrease
	Power	Increase → decrease	Decrease	Increase
Initial value	Pressure	73 kg/cm ² a	73 kg/cm ² a	73 kg/cm ² a
	Flow rate	55 t/h	55 t/h	55 t/h
	Inlet enthalpy	abt. 293 kcal/kg	abt. 293 kcal/kg	abt. 293 kcal/kg
	Power	6 MW	6 MW	6 MW
Change range	Pressure	1.0 → 1.10	Constant	Constant
	Flow rate	1.0 → 0.3	1.0 → 0.6	Constant
	Inlet enthalpy	Constant	Constant	1.0 → 0.97
	Power	1.0 → 1.15	1.0 → 0.75	1.0 → 1.15
Sub-total		1 data	1 data	1 data

within a 1% error. These results proved that the design method could closely predict the total pressure loss at the heat generating part in both single and two phase flows, irrespective of the test assembly shape.

(d) Void characteristic test

This test was carried out with test assemblies (II) and (IV).

1) **Test method** Void Fraction was measured by means of an X-ray densitometer at the three windows in the outlet piping of test assembly, where the simulated high-temperature high-pressure steady two-phase fluid flowed. Average void fraction was determined from the three measured void fractions by:

- ① Functional approximation
- ② Center line symmetrical division
- ③ Axial symmetrical division

④ Straight-line interpolation

2) **Test conditions** The test range was as follows.

Test assembly: New 8×8 fuel assembly.

Axial power distribution: Cosine distribution & inlet peak distribution.

Radial power distribution: Distribution A, simulating power distribution in early fuel burnup stage.

Pressure: 73 and 88 kg/cm² a

Flow rate: 20~65 t/h

Inlet subcooling: 12~30 kcal/kg

Outlet quality: 0.07~0.25.

One hundred and nineteen bits of void fraction data were collected within the above range.

Test conditions are shown in Table 1・12.

3) **Test results** Figures 1・25 and 1・26 show

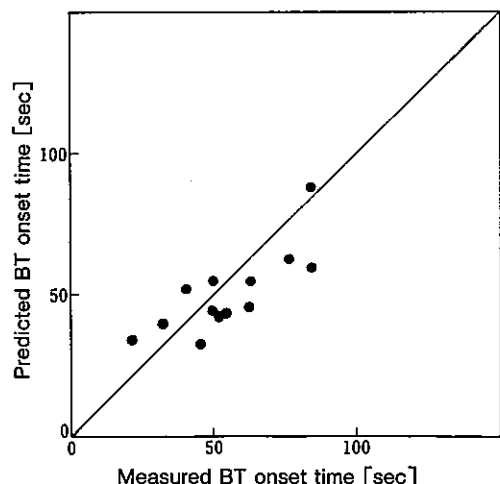


Fig. 1 · 16 Result of loss of feedwater heating test for test assembly (III) (Test data vs. prediction)

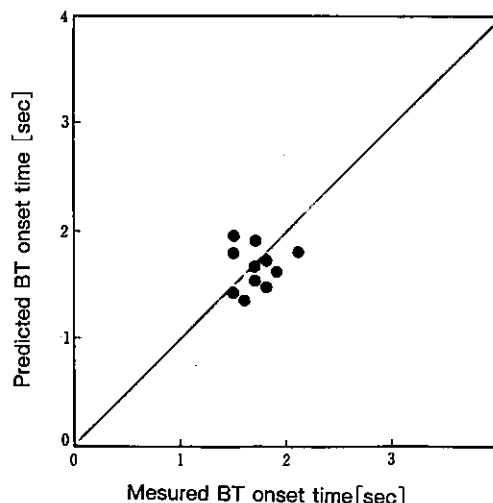


Fig. 1 · 19 Result of turbine trip without bypass test for test assembly (III) (Test data vs. prediction)

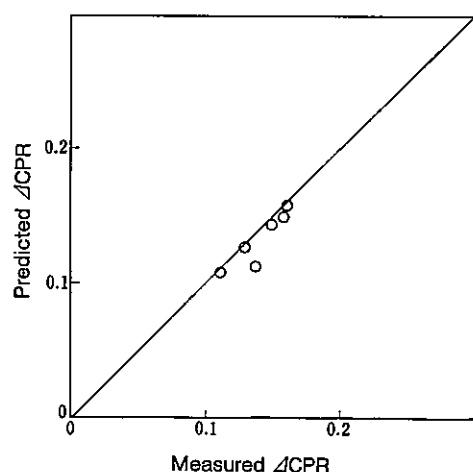


Fig. 1 · 17 Result of loss of feedwater heating test for test assembly (III) (Test data vs. prediction)

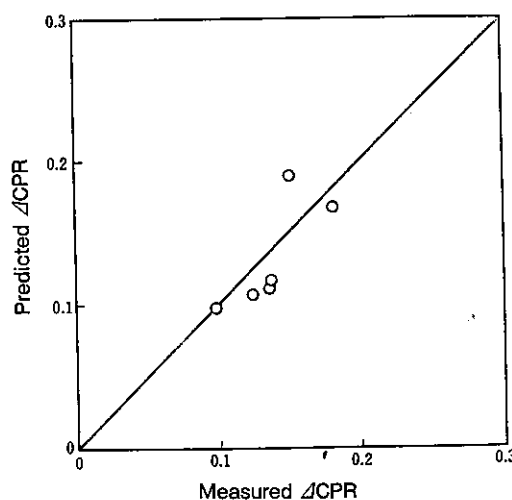


Fig. 1 · 20 Result of turbine trip without bypass test for test assembly (III) (Test data vs. prediction)

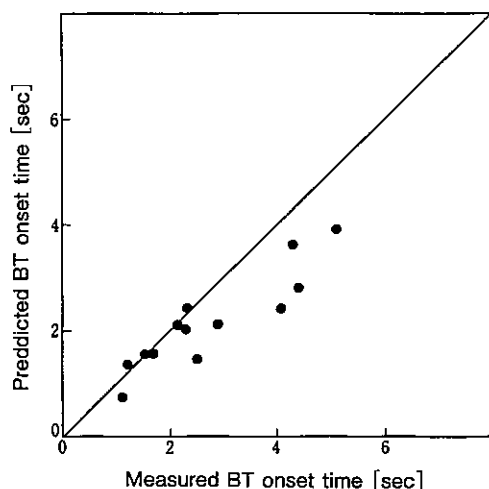


Fig. 1 · 18 Result of recirculation pump trip test for test assembly (III) (Test data vs. prediction)

a comparison of the calculated value by the design method with the measured average void fraction under the simulated rated reactor operation conditions, with the axial power distribution being the cosine distribution and inlet peak distribution. The comparison indicates that the design value tends to overestimate the measured value in the low void fraction range (less than 50%) and underestimate it in a higher range. Any quantitative evaluation is difficult because the average void fraction differs with the estimation methods used. However, regarding the functional approximation (plot □) method, that takes nearly an average of those given by the other four methods, the design value differs from the measured value by about 3% and 1% (absolute void fraction values) at maximum in the low void fraction range with the axial power distribution being cosine and inlet peak distributions, respectively, and by about -5% at maximum in the higher range.

Table 1・10 Condition for single phase flow test for test assembly (I) (II)

Pressure [kg/cm ² a]	Inlet Sub- cooling [kcal/kg]	Flow [t/h]											
		20	25	30	35	40	45	50	55	60	65	70	75
73	12	○	○	○	○	○	○	○	○	○	○	○	○
Atmo- spheric Pressure	Normal Temper- ature	○	○	○	○	○	○	○	○	○	○	○	○

Table 1・11 Condition for two phase flow test for test assembly (I) (II)

Pressure [kg/cm ² a]	Flow [t/h]	Inlet Sub- cooling [kcal/kg]	Outlet Quality				
			0.07	0.10	0.15	0.20	0.25
73	20	12	○	○	○	○	○
	45	"	○	○	○	○	○
	55	"	○	○	○	○	○
	65	"	○	○	○	○	○
88	20	12	○		○		○
	45	"	○		○		○
	55	"	○		○		○
	65	"	○		○		○

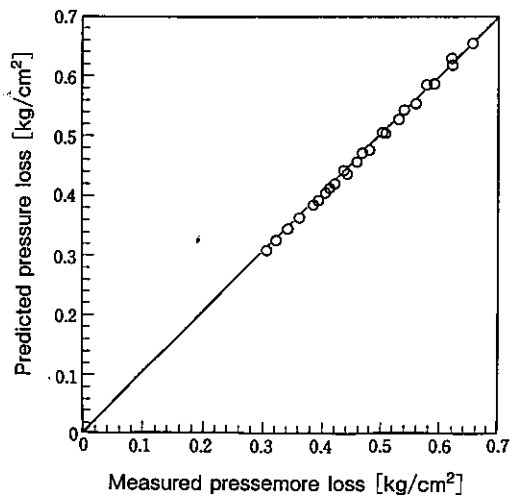


Fig. 1・21 Result of single phase flow test for test assembly (I)

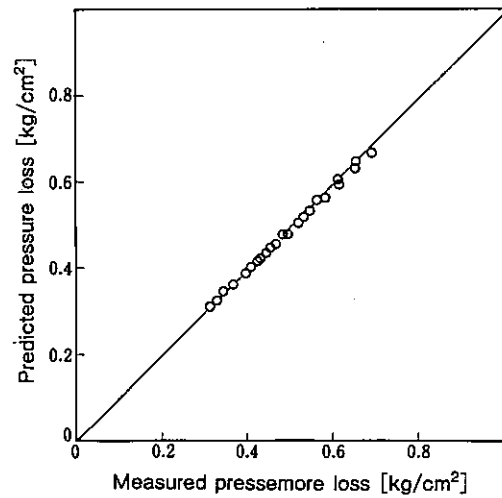


Fig. 1・23 Result of single phase flow test for test assembly (II)

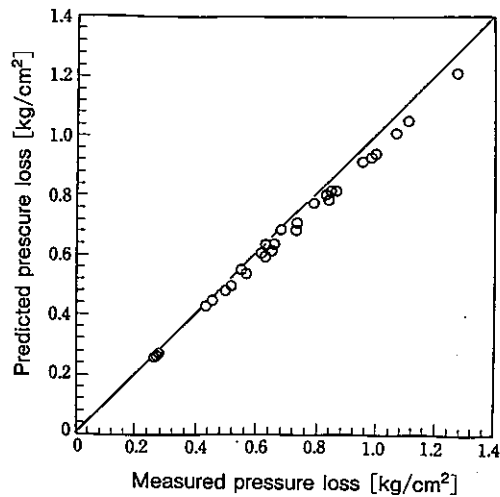


Fig. 1・22 Result of two phase flow test for test assembly (I)

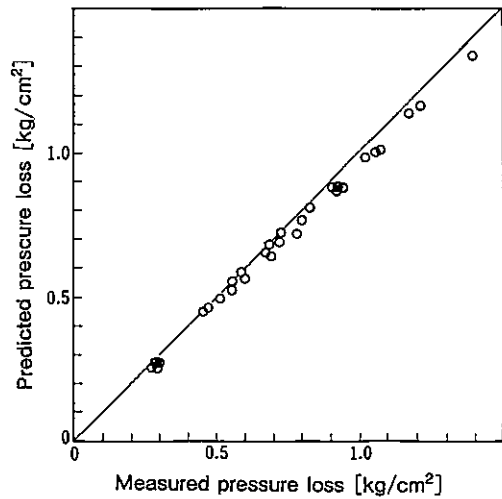


Fig. 1・24 Result of two phase flow test for test assembly (II)

Table 1・12 Condition of void characteristic test for test assembly (II) (IV)

Pressure [kg/cm ² a]	Flow [t/h]	Inlet subcooling [kcal/kg]	Quality				
			0.07	0.10	0.15	0.20	0.25
73	20	12	○*	○*	○*	○*	◎*
		20	○	○	○	○	○
		30	○*	○*	○*	○*	○*
	45	12	○	○	○	○	○
		20	○	○	○	○	○
		30	○	○	○	○	○
	55	12	○*	○*	○*	○*	◎*
		20	○	○	○	○	○
		30	○*	○*	○*	○*	○*
	65	12	○*	○*	○*	○*	○*
		20	○	○	○	○	○
		30	○*	○*	○*	○*	○*
88	20	12	○		○		○
		20	○		○		○
	45	12	○		○		○
		20	○		○		○
	55	12	○		○		◎
		20	○		○		○
	65	12	○		○		○
		20	○		○		○

○* : Test carried out with both (II) and (IV).
○ : Test carried out with only (II).
◎ : Reproducibility test

From these results, the design method was proved to rather accurately predict the average void fraction at the outlet piping of test assembly, irrespective of different axial power distributions (cosine or inlet peak distribution), considering the estimated measurement error and that the large caliber piping (180 mm) had been measured with 3 beams. **Figures 1・27 and 1・28** show comparison of the test results with typical published void fraction correlations (Armad, Bankoff,

Zuber & Findlay). The comparison shows a worse agreement than that with the design method in **Figures. 1・25 and 1・26**, the design method is confirmed to have more accurately predicted the test results than the published correlations.

[6] Summary

The critical power data under steady and transient conditions, flow characteristic data and void characteristic data for the mock-up BWR fuel assemblies, could be obtained by the proving test carried out from April, 1978 to March, 1987. Detailed analysis and evaluation of these data proved the adequacy of the present reactor core thermal hydraulic design method.

Results of this proving test can be summarized as follow.

(a) Reliability of test data

This proving test was carried out by the test facility (maximum thermal loading test facility) capable of realizing thermal hydraulic conditions during BWR operation using test assemblies simulating the shape and heat generation distribution of the actual fuel assembly by electric heating.

The following four kinds of test data, necessary for proving the adequacy of the reactor core thermal hydraulic design method, were obtained by the test.

- 1) 592 bits of data for the maximum thermal loading test under steady conditions
- 2) 63 bits of data for the maximum thermal loading test under transient conditions
- 3) 118 bits of data for the flow characteristic test
- 4) 119 bits of data for the void characteristic test

The test facility operated favorably and the data were collected, after carefully calibrating the measuring instruments. Reproducibility tests were conducted to confirm that the data were properly collected. The test assembly was inspected after tests, but no particular changes from the condition before the test were found.

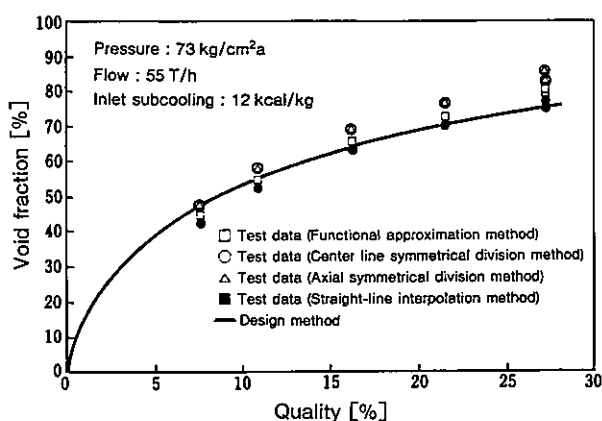


Fig. 1・25 Result of void characteristic test for test assembly (II)

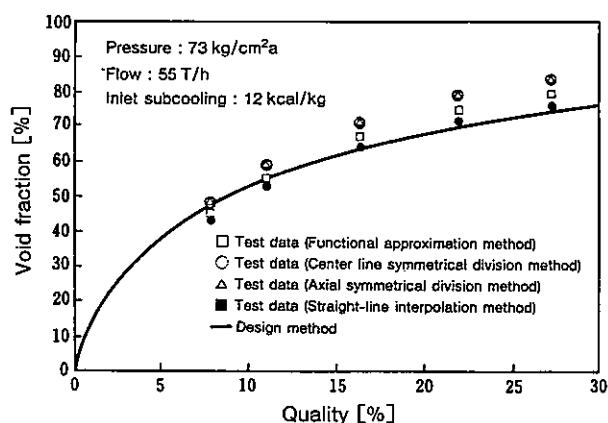


Fig. 1・26 Result of void characteristic test for test assembly (IV)

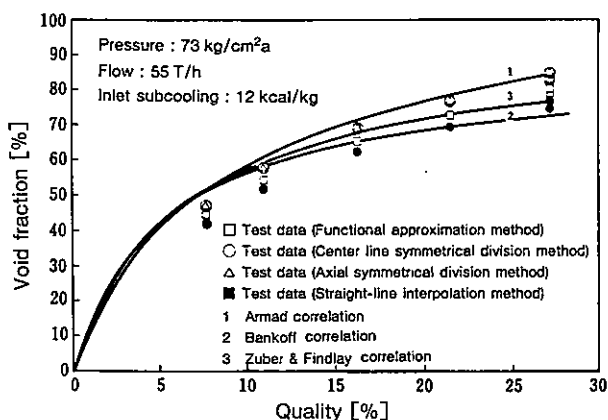


Fig. 1-27 Comparison of measured void fraction with typical published void fraction correlations (test assembly (II))

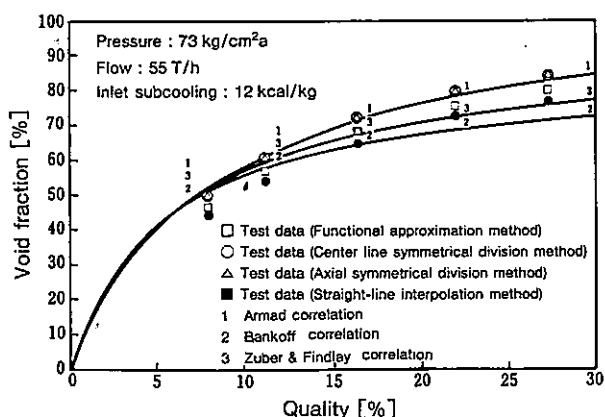


Fig. 1-28 Comparison of measured void fraction with typical published void fraction correlations (test assembly (IV))

The obtained data agreed well with the published data in the qualitative trends, such as influences of various parameters, etc. These results have proved the high reliability of the data obtained from this test.

(b) Reliability of BWR reactor core thermal hydraulic design method

1) Maximum thermal loading test under steady conditions The calculated value by GEXL correlation agreed well with the measured value (average ECPR was 1.001), with its standard deviation being 3.4%. This value was less than the standard deviation of 3.6% considered in applying GEXL correlation to the present reactor core design. Further, the heater rod temperature was raised after boiling transition condition to evaluate the judgement criteria for critical power (heater rod surface temperature difference : 14°C). This test verified that the critical power corresponding to the heater rod surface temperature difference (14°C) was conservative.

2) Maximum thermal loading test under transient

conditions Test were carried out using the test assembly with axial power distribution of cosine distribution and inlet peak distribution, simulating the change in BWR analysis parameters. As a result, the boiling transition was confirmed not to occur. Further, the analysis value determined, by the single channel thermal hydraulic analysis code, was compared with the measured value regarding the boiling transition onset time and Δ CPR. This confirmed the good agreement of the analysis value with the measured value within the GEXL correlation precision range, and the reliability of the present thermal hydraulic evaluation method for transient.

3) Flow characteristic test The measured total pressure loss in the test assembly was compared with the calculated value by the design method. Results showed that they closely agreed with each other, irrespective of the test assembly shape (new 8×8 retrofit fuel assembly or new 8×8 fuel assembly), and the reliability of the present thermal hydraulic design method was verified.

4) Void characteristic test The measured void fraction at the outlet piping for the test assembly was compared with the calculated value by the design method. It was ascertained that the design method could predict the void fraction at outlet piping with accuracy, irrespective of different axial power distributions (cosine distribution or inlet peak distribution), and the reliability of the present thermal hydraulic design method was confirmed.

2. PWR Fuel Assembly Maximum Thermal Loading Test

[1] Outline of the Test

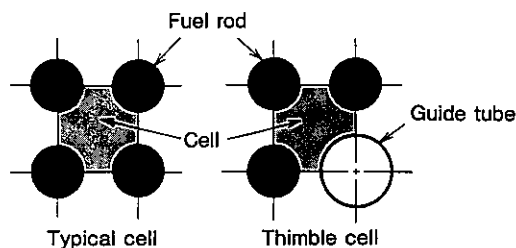
The PWR fuel assembly maximum thermal loading test is to measure the maximum thermal loading (DNB Heat Flux, DNB : Departure from Nucleate Boiling), etc. under various conditions using the high-temperature and high-pressure out-of-pile thermal hydraulic test facilities and mock-up fuel assemblies for the purpose of confirming the reliability of PWR core thermal hydraulic design method and the thermal margin of PWR fuel assemblies.

Main specifications of the test facilities are shown in Table 1-13. This facilities was constructed in Takasago Engineering Laboratory.

This test is to be performed for a mock-up part (5×5 or 6×6 arranged in square) of the PWR 17×17 type fuel assembly. The fundamental design of the entire proving test was performed in fiscal year 1978 and its detailed design in 1983. Later the construction of the test facilities was completed during fiscal year 1984 and 1987, and the test started from fiscal 1988.

Table 1・13 Specifications for PWR fuel assembly maximum thermal loading test facilities

Item	Specification
Maximum heating power [MW]	10
Maximum coolant flow rate [t/h]	60
Maximum coolant pressure [kg/cm ² a]	169



● Definition

Typical cell : Cell surrounded by four fuel rods

Thimble cell : Cell surrounded by three fuel rods and one guide tube for control rod or In-core monitor

Fig. 1・29 Typical cell and thimble cell

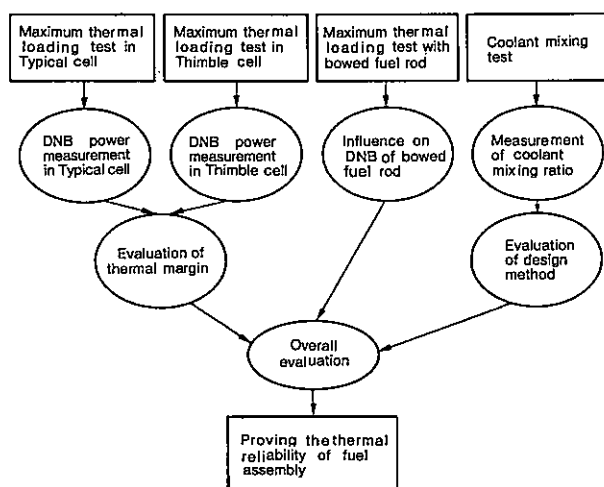


Fig 1・30 Evaluation procedure for PWR fuel assembly maximum thermal loading test

[2] Contents of the Test

The following items are to be verified.

(a) Maximum thermal loading test in typical cell

The maximum thermal loading test is performed under the same high-temperature and high-pressure conditions as in the power reactor, using the test bundles simulated actual typical cell (Figure 1・29) of the fuel assembly. The thermal margin in the normal operating conditions and the possible transient state of the power reactor is obtained and evaluated.

(b) Maximum thermal loading test in thimble cell

The maximum thermal loading test is performed under the same conditions as in the power reactor described above, using the test bundles simulated actual thimble cell (Figure 1・29) of the fuel assembly. The thermal margin in the normal operating conditions and the possible transient state is obtained and evaluated.

(c) Maximum thermal loading test with bowed fuel rod

The effect of bowed fuel rod is verified by maximum thermal loading test, using the test bundles including bowed fuel rod.

Table 1・14 Main specifications for test bundles simulated PWR fuel assembly

Item	Specification
Heater rod array arrangement	5×5 (6×6)
Outer Diameter of heater rod [mm]	9.50
Outer Diameter of guide thimble [mm]	12.24
Pitch of heater rod center [mm]	12.60
Effective length of rod [mm]	3660
Number of heater rods	
① In the case of Typical Cell	25 (36)
② In the case of Thimble Cell	24 (35)

Figures in parenthesis show the number of heater rods in the case of 6×6 heater rod array arrangement.

Table 1・15 Type of test bundles for PWR

Test bundle No.	1	2	3	4	5	6	7	8	9	10	11	12
Item												
Rod array arrangement	5×5	5×5	6×6	5×5	5×5	6×6	5×5	5×5	5×5	5×5	5×5	5×5
No. of grids	9	9	9	9	9	9	9	9	9	9	9	9
Type of cell	—	Typical	Typical	Typical	Typical	Thimble	Thimble	Thimble	Thimble	Thimble	Typical	Thimble
Axial power distribution	Uniform	Uniform	Uniform	Cosine	Cosine	Uniform	Cosine	Cosine	Cosine	Cosine	Cosine	Cosine
Type of test	Mixing test	Steady state test							Test with bowed fuel rod		Transient test	

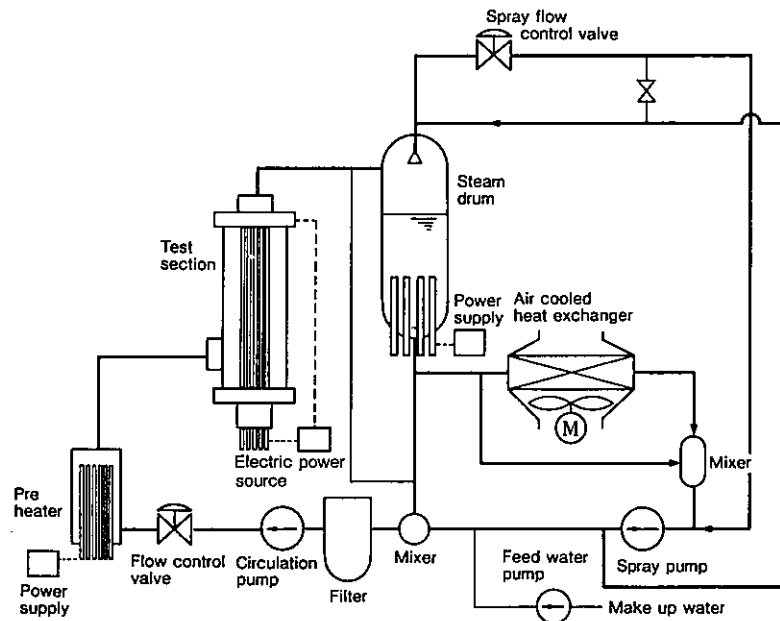


Fig. 1-31 System diagram for PWR maximum thermal loading test facilities

(d) Coolant mixing test

The mixing ratio for the coolant under high-temperature and high-pressure is measured and evaluated, using the test bundle simulated the actual fuel assembly.

The evaluation procedure for each test is shown in Figure 1-30.

[3] Outline of Test bundles

Outline of the test bundles is mentioned below.

(a) The fuel assembly under investigation is 17×17 type with 9 grids, but the test bundles are 5×5 or 6×6 type simulated fuel assemblies arranged in square.

(b) The test bundles are fabricated for typical cell and for thimble cell to make them represent the DNB characteristics of fuel assembly.

(c) A cosine distribution is selected as the representative axial power distribution of test bundles considering the design and operating conditions of PWR, but test bundles for uniform distribution are also fabricated to confirm the influences of the axial power distribution.

(d) The occurrence of DNB is detected by the thermocouples attached to the inside of the heater rod.

Main specifications of the test bundles are shown in Table 1-14 and the type of test bundles in Table 1-15.

[4] Outline of Test Facilities

This test facilities are a high-temperature and high-pressure thermal hydraulic test facilities which simulates the thermal hydraulic conditions in PWR fuel assemblies by direct electric heating of the test bundles

- ① Test section
- ② Steam drum
- ③ Circulation pump
- ④ Filter
- ⑤ Flow control section
- ⑥ Preheater
- ⑦ Spray pump
- ⑧ Spray flow control section

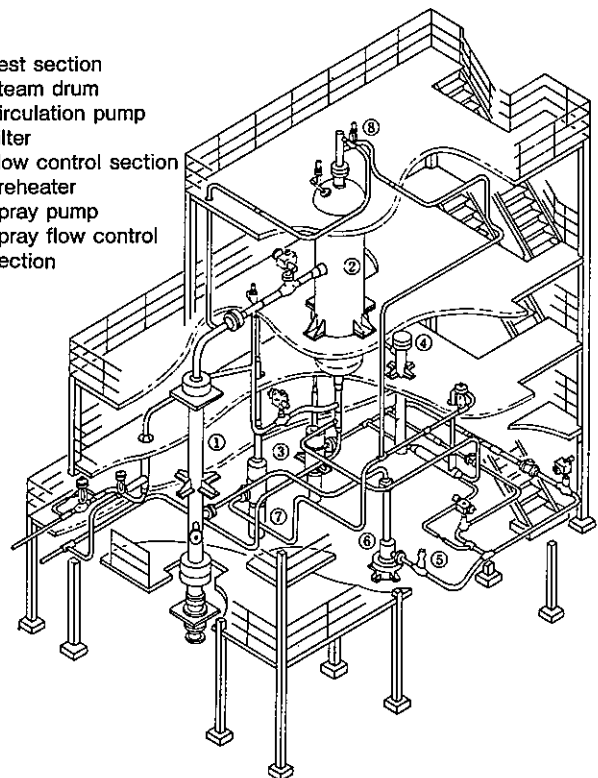


Fig. 1-32 Bird's eye view of PWR fuel assembly maximum thermal loading test facilities

(simulated fuel assembly) installed into the test section. This facilities consist of the main circulating loop facilities including the test section, the auxiliary system, electric power source system, the instrument and control system and auxiliary equipment.

A system diagram of the facilities is shown in **Figure 1 · 31** and a bird's-eye view of the test facilities is shown in **Figure 1 · 32**. The specifications for the main components of the test facilities are shown in **Table 1 · 16**.

(a) Main circulating loop facilities

The main circulating loop facilities supply high-temperature and high-pressure water to the test section installed the simulated fuel assemblies at the temperature, pressure and flow rate decided according to the operation conditions. This facilities consist of the test section, the circulating system and the cooling system. Main components are shown below.

1) Test section The test section consists of a internal structure for housing the simulated fuel assembly, a test pressure vessel for holding it and accessories to the test section such as upper and lower electrodes. The test vessel can install the simulated fuel assemblies of 6×6 type with a length up to 4.3 m.

2) Steam drum The steam drum is a pressure vessel provided with a heater and a water spray nozzle and having the following functions.

- The steam drum is used as the pressurizer when increasing the temperature and pressure. It raises the pressure in the main circulating loop prior to raising the temperature.
- It controls the pressure in the main circulating loop by adjusting the flow rate of spray water and the heater power.
- It stabilizes the operating conditions by absorbing the change in the volume of stored water caused by the

Table 1 · 16 Specifications of main components in PWR fuel assembly maximum thermal loading test facilities

Equipment		Components	Specifications	Equipment		Components	Specifications
	Power source	Test section	Type : Vertical cylinder Design pressure : 195 kg/cm ² G Design temperature : 362°C		Rectifier transformer (2 units)	Type : Sending oil outdoor water-cooled type with conservator Rated frequency : 60 Hz Phase : AC side 2×3 phase/DC side 2×6 phase Rated capacity : 2×3240 kVA/2×4585 kVA Voltage : 6.6 kV/235 V	
		Steam drum	Type : Vertical cylinder Design pressure : 195 kg/cm ² G Design temperature : 362°C Total heat generation : 750 kW				
		Preheater	Type : Sheath heater heating system Design pressure : 195 kg/cm ² G Design temperature : 362°C Total heat generation : 450 kW				
		Air-cooled heat exchanger	Type : Lateral forced ventilation type Design pressure : 195 kg/cm ² G Design temperature : 362°C Maximum heat exchange : 13.5 MW			Thyristor rectifier (2 units) Type : Outdoor closed independent type Rated frequency : 60 Hz Rated DC power : 2×2646 kW Rectification circ it system : Double star type 2×6 phase with inter-phase reactor	
	Instrumentation and control	Circulation pump	Type : Canned motor pump Design pressure : 195 kg/cm ² G Design temperature : 362°C Flow rate : 100 m ³ /h Lift : 200 m		Instrumentation supervision panel	Type : Indoor independent type (rack type) Main components : Linear coder, maximum thermal load detection unit, maximum thermal load detection value setter, maximum thermal load display lamp, temperature indicator, temperature compensation converter unit.	
		Spray pump	Type : Canned motor pump Design pressure : 195 kg/cm ² G Design temperature : 362°C Flow rate : 90 m ³ /h Lift : 210 m			Type : Indoor closed independent type Main components : Graphic board, digital display, indicator, recorder, alarm display lamp.	
						Central supervision panel Type : Indoor control desk type Main components : operation switch, indicator, display lamp.	
						Process I/O controller Number of channels : 256 Sampling cycle : 0.1 sec/256 channels	

external turbulence during the test or changes in the test conditions.

- It secures the heater capacity that can respond to the transient test.

3) Preheater The preheater makes fine adjustment of the inlet temperature of the coolant in the test section through the control of sheath heater power.

4) Air-cooled heat exchanger The temperature and pressure of the main circulating loop continue to rise as the test bundle, etc. are heated, so the air-cooled heat exchanger is provided to maintain the desired temperature and pressure by cooling the circulating water. The control system adopted is a system which controls the water temperature at the heat exchanger exit by changing the cooling air flow rate by adjusting the fan blade angle or louver.

5) Circulation pump The circulation pump is provided in the circulating system and functions to supply the test vessel with the necessary volume of circulation water. The flow rate is controlled by adjusting the opening degrees of the flow rate control valve and the electric valve for changing the flow rate measurement line.

6) Spray pump The spray pump is installed in the cooling system and supplies the necessary volume of water to the spray line and the circulating loop mixer. The flow rate of the spray line is adjusted by a flow rate control valve and an electric valve, and its flow rate to the circulating loop mixer is adjusted by a flow rate control valve.

(b) Auxiliary system

The auxiliary system functions as the auxiliary equipment of the main circulating loop facilities. It performs the work such as water supply to the main circulating loop, cooling the test section electrodes, supply of cooling water to the heat exchanger, supply of air for instrumentation, supply of nitrogen gas, calibration of the flow meter of the main circulating loop, etc.

(c) Electric power source system

The electric power source system consists of the high-voltage electricity reception/delivery system, the power source for heating the test bundles, the heater power source, and the power source for common use. The system was enlarged to cover shortage by making best use of the existing electric power source system.

(d) Instrument and control system

The instrument and control system sets up various control values for the process line and control it to realize the desired operation conditions, collects the measurement data and supervises various operation data for the process line. This system consists of the instrument and control system of the test section, the process instrument and control system and the data processing equipment.

(e) Auxiliary equipment

The auxiliary equipment is provided for handling the simulated fuel assemblies and used for loading them in the test section, replacing and storing them.

[5] Proving Test

This proving test includes a coolant mixing test, maximum thermal loading tests under steady conditions and transient conditions using a mock-up part (5×5 or 6×6 arranged square) simulating A and B type 17×17 fuel assemblies for the purpose of proving the reliability with regard to the thermal performance of PWR fuel assemblies. The test is being performed during the period from fiscal 1988 to 1989. In fiscal 1988, a coolant mixing test and a maximum thermal loading test under steady conditions were made on test specimens Nos. 1 through 7.

(a) Mixing test

The mixing ratio of coolant under the high-temperature and high-pressure conditions is measured at the coolant outlet using the test bundles simulating the fuel assemblies used in the actual reactor.

The mixing ratio of coolant obtained from this test is analyzed and evaluated to confirm that thermal diffusion coefficient used in the present design is sufficiently at the conservative side and to decide the thermal diffusion coefficient used for the analysis of the maximum thermal loading test data. The thermal diffusion coefficient is the coefficient used for evaluating the heat mixing ratio of the adjacent channels in the thermal hydraulic calculation code for PWR core. The coefficient depends on geometry of fuel assemblies, etc.

The coolant mixing test was made on test bundles No. 1. Table 1-15 shows its specifications.

2) Test Conditions Table 1-17 shows the test parameter conditions in the mixing test.

3) Test Results Fig. 1-33 shows the thermal diffusion coefficient determined through the analysis of the test results obtained. It was confirmed based on the test results that the thermal diffusion coefficient has no dependency on Reynolds number.

It was further confirmed that the thermal diffusion

Table 1-17 Condition of mixing test

Flow G [10 ⁶ Kg/m ² · hr] Pressure [Kg/m ² ·a]	0.5	1.2	2	5	11	17
50	○	○	○	○	○	○
100	○	○	○	○	○	○
150	○	○	○	○	○	○
169	○	○	○	○	○	○

coefficient presently used in the design is conservative enough.

(b) Maximum thermal loading test under steady conditions

In PWR, the mode of heat transfer from the fuel rod surface to the coolant is forced convection or nucleate boiling heat transfer, when the heat flux increases abnormally or the pressure or flow rate lowers abnormally, the heat transfer mode shifts from nucleate boiling to film boiling. When the heat transfer mode has

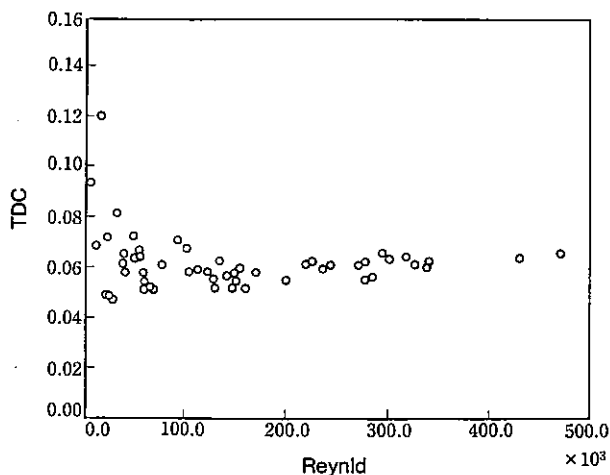
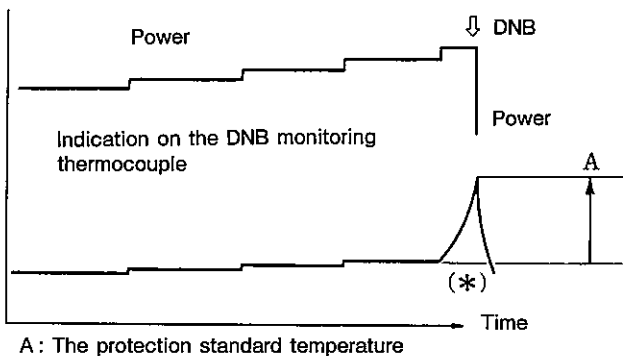


Fig. 1 · 33 Result of measured TDC for No. 1 test



A : The protection standard temperature

Fig. 1 · 34 Relations between the output rise leading the occurrence of DNB and the indications on the DNB monitoring thermocouple.

Table 1 · 18 Condition of DNB test

Flow G [10 ⁶ Kg/m ² · hr]	2	5	8	11	14	17
Pressure [Kg/m ² ·a]						
50	○	○	—	○	—	—
75	○	○	○	○	—	—
100	○	○	○	○	○	○
125	○	○	—	○	○	○
150	○	○	○	○	○	○
169	—	○	—	○	—	○

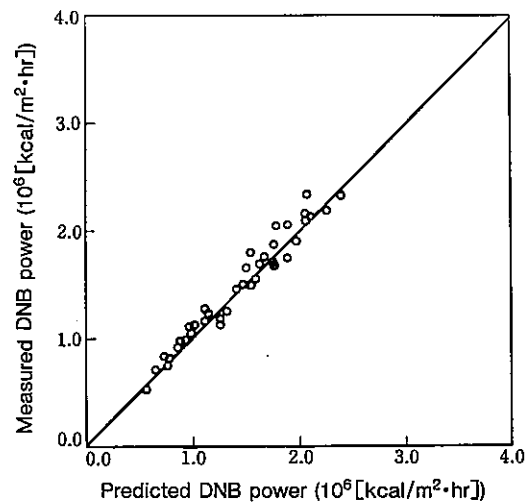


Fig. 1 · 35 DNB post data VS. MIRC-1 prediction for No. 2 test.

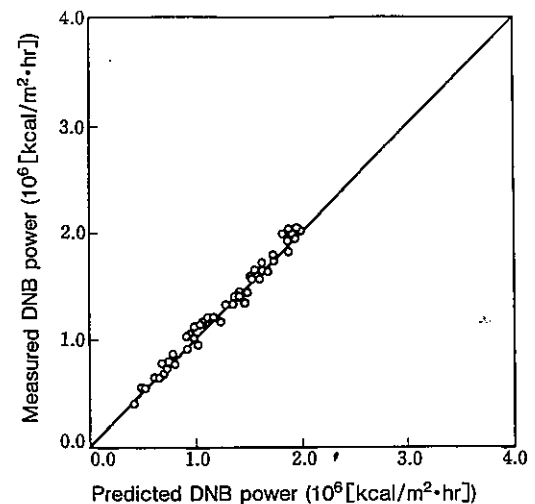


Fig. 1 · 36 DNB power data VS. MIRC-1 prediction for No. 3 test.

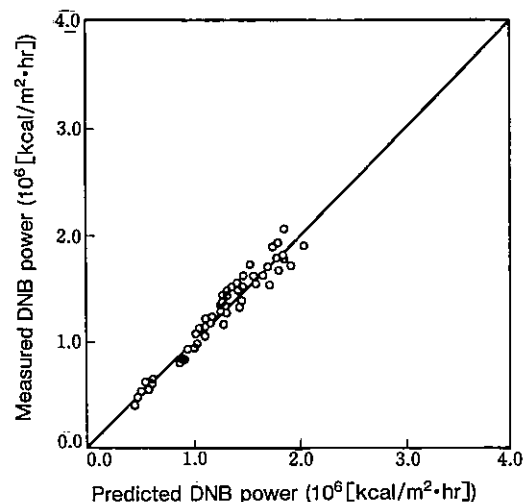


Fig. 1 · 37 DNB power data VS. MIRC-1 prediction for No. 4 test.

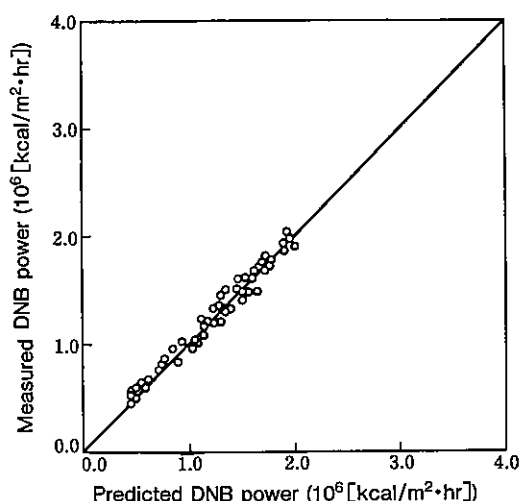


Fig. 1 · 38 DNB power data VS. NFI-1 prediction for No. 5 test.

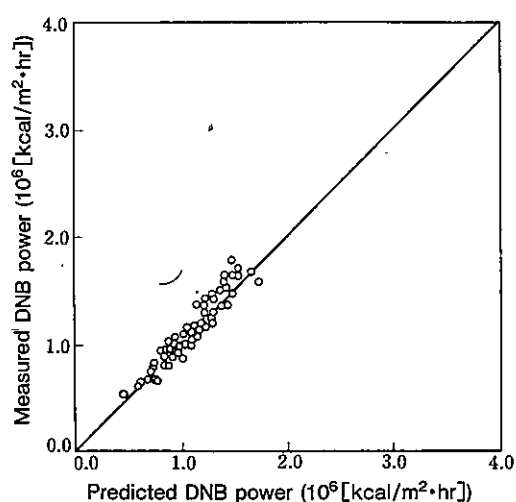


Fig. 1 · 39 DNB power data VS. NFI-1 prediction for No. 6 test.

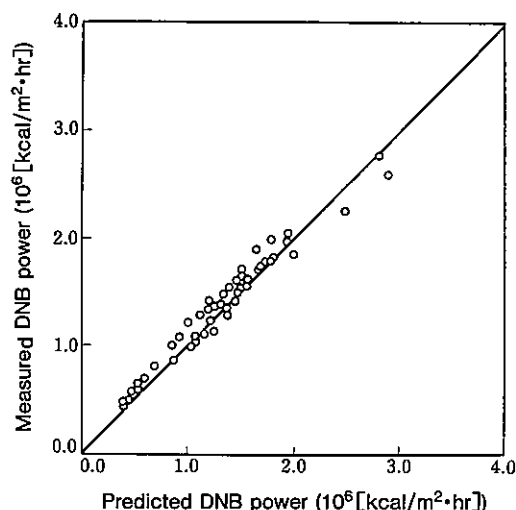


Fig. 1 · 40 DNB power data VS. NFI-1 prediction for No. 7 test.

shifted to film boiling, the heat transfer coefficient from fuel rod surface to coolant lowers rapidly and the clad temperature rises rapidly. The point where this heat transfer mode shifts is defined as the maximum thermal loading. A maximum thermal loading test is performed under broad test conditions including the coolant conditions of the actual reactor using the test bundles simulating a normal channel and the channel including thimbles, to evaluate the thermal allowance during normal operation. The test bundles Nos. 2 to 4 simulate A type 17×17 fuel assemblies and the test bundles Nos. 5 to 7 B type 17×17 fuel assemblies. Table 1 · 15 shows the specifications of these test bundles.

① Test method In the maximum thermal loading test under steady conditions, the system pressure, flow rate and the inlet temperature are set up according to the conditions, and then the power input to the test bundles is increased to reach DNB. The power input to the test bundles is increased up to the vicinity of DNB heat flux based on the preliminary analysis results and experience, followed by gradually increasing the heated rods power by fine steps. When DNB occurs, the temperature of the thermocouples attached to the positions where the occurrence of DNB is expected begins to rapidly rise. That is, the occurrence of DNB is confirmed by this rapid temperature rise indicated by the thermocouple recorder (see Fig. 1 · 34). As regards the method of reducing the output after the occurrence of DNB, the output is reduced when the temperature of the thermocouple has risen 11°C or more in the even distribution (10°C or more in the cosine distribution) immediately before the occurrence of DNB. Incidentally, the protection standard for the test bundles is that they shall automatically trip when the thermocouple temperature has risen 15°C.

② Test conditions Table 1 · 18 shows the test conditions in the maximum thermal loading test under steady conditions.

③ Test results Figs. 1 · 35 to 1 · 37 show the results of comparing the measured values with the forecast values calculated from the A type test bundles data using the MIRC-1 equation. Figs. 1 · 38 to 1 · 40 shows the same results with the forecast values calculated from the B type test bundles by means of the NFI-1 equation. It was confirmed from these figures that the MIRC-1*1 and NFI-1*2 equations used in the design are correlating well with the data within the applicable range.

*1 : MIRC-1 equation was prepared by Mitsubishi so that the equation could have correlations with the data of DNB test simulating the WH type PWR fuel conducted by Columbia University, the U. S. A.

*2 : NFI-1 equation was prepared by Nuclear Fuel Industries, Inc. so that the equation could have correlations with the data of DNB test simulating WH type and B type PWR fuel conducted by Columbia University, the U. S. A.

II. VOID FRACTION MEASUREMENT TEST IN FUEL ROD BUNDLES

The reliability of the fuel assemblies used in nuclear power plants is the basic element to secure the safety of reactor, and the fuel assemblies are required to keep its integrity even under the thermally critical conditions such as abnormal transient conditions, besides during normal operation.

In light water reactors (LWR), electricity is generated by the turbine generator, rotated by the steam generated by heating the coolant with the heat produced by nuclear fission of fuel assemblies. Because the coolant functions as the moderator of neutron in addition to remove the heat generated from fuel assemblies, the behaviors of void (steam bubbles) in the coolant largely affect the nuclear and thermal characteristics of the reactor, and are one of the most important factors in the core-fuel design.

In the present LWR core-fuel design, various events are analyzed using the void correlations based on a single exothermic tube test. In order to further improve the reliability of the core-fuel design, it is necessary to grasp the void behavior within fuel assemblies in detail.

So, this test is intended to prove the reliability of the void correlations and fill up the void data base and improve the reliability of the analysis method, by making clear the detailed void behaviors in fuel rod bundles composed of plural fuel rods in the fuel assemblies used in the actual reactor.

In this test, one of the electrically heated mock-up fuel assemblies (test assemblies) that can simulate the same heated conditions as in the fuel assemblies of the actual reactor is loaded in the thermal hydraulic test facility which can simulate the same high-temperature and high-pressure conditions as in an actual reactor, in order to measure the void fraction distributions in the coolant that flows through the fuel assembly. The proving test is carried out by simulating pressure, coolant flow rate and power of fuel elements, under the normal operating steady conditions of BWR and during the transient and accidental conditions of BWR and PWR.

1. Void Measurement Test in BWR Fuel Assembly

[1] Outline of the Test

This test is intended to prove the reliability of the BWR nuclear thermal hydraulic design method by measuring the void fraction distribution in the fuel assembly using the thermal hydraulic test facility that can simulate the same high-temperature and high-pressure conditions as in BWR.

For this test, the BWR fuel assembly maximum thermal loading test facility at Isogo Engineering Laboratory that has already completed the proving test is partially remodelled, and electrically heated full-size 8X8 mock-up fuel assemblies (test assemblies) that can simulate the same heated conditions as in the actual reactor are fabricated to perform the proving test using them.

With regard to this proving test, its basic design was carried out and the manufacture of the test facility started in fiscal year 1987.

[2] Contents of the Test

The following items will be tested.

(a) Void measurement test under steady conditions

The void fraction distribution in the coolant flowing through the fuel assembly is measured using the parameters of pressure, coolant flow rate, power, power distribution, etc. which affect the void characteristics.

The range of the test parameters basically covers the range of actual reactor operation as follows.

1) **Pressure: 10-88 [kg/cm²]** The pressure is set up at 10-88 kg/cm² so as to cover from the hot standby and normal operating conditions to the transient analysis and evaluation range (63-88 kg/cm²).

2) **Coolant flow rate: 10-70 [t/h]** The flow rate is set up at 10-70 t/h, considering the transient analysis and evaluation range that is about 15-60 t/h.

3) **Exit quality: 0-25 [%]** The exit quality is set up at 0-25% so as to cover the exit quality (about 20%) of actual reactor rated operation, in order to simulate the exit quality changes that correspond to the power changes.

4) **Power distribution** Concerning the axial power distribution, a cosine distribution is basically adopted as the typical power distribution in the actual reactor. Test is also performed on the uniform distribution, the inlet and outlet peak distributions in order to confirm the dependency of the void fraction on the axial power distribution.

The radial power distributions of the test assembly are simulated the beginning of life of fuel in an example of BWR fuel design which has the largest local peaking factor.

(b) Void measurement test under transient conditions

The change in the void fraction with the passage of time of the coolant flowing through the fuel assembly is measured, simulating changes in the pressure, flow rate, power, etc. with the passage of time under the

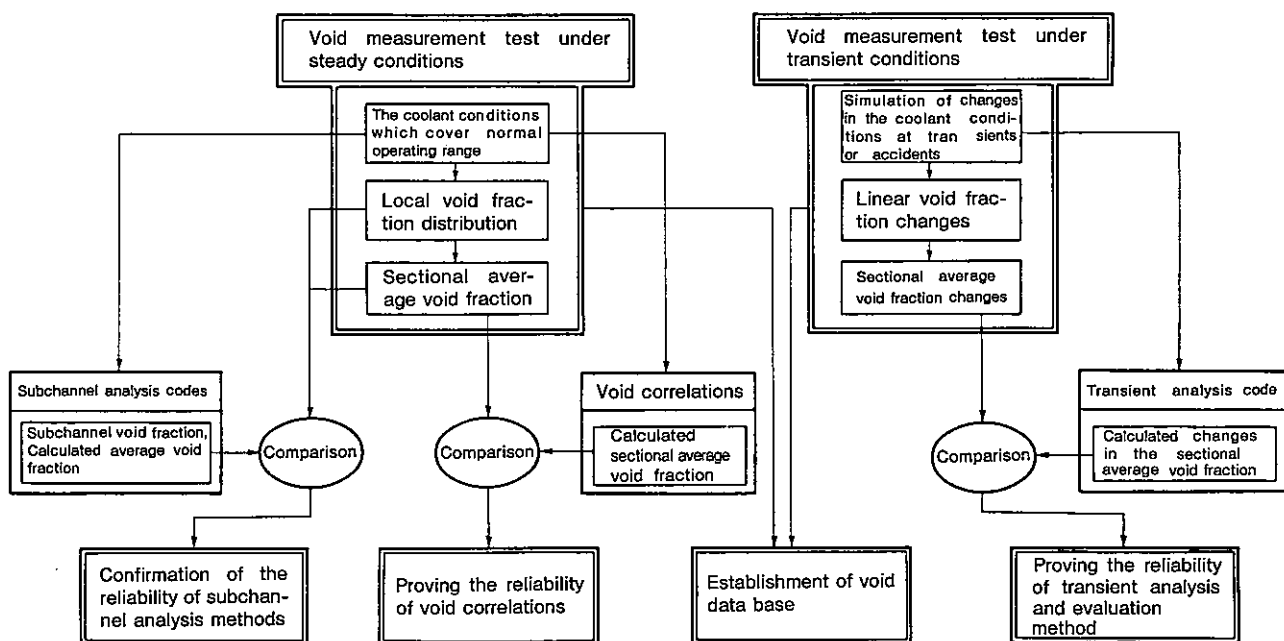


Fig. 2 · 1 Evaluation procedure for BWR fuel assembly void measurement test

typical phenomena among the abnormal transients and accidents during operation of a reactor.

The following four phenomena are tested, in which the core void fraction remarkably changes and the void behaviors are important in connection with the dynamic neutron characteristics under the abnormal transients and accidents during operation of BWR.

- (1) The generator load rejection with bypass failure
- (2) The recirculation pump trip

- (3) The recirculation pump seizure
- (4) The pressure controller failure

Among the phenomena in which the core void fraction reduces, the generator load rejection with bypass failure is selected as the phenomenon where transient changes are remarkable.

The phenomena where the core void fraction increases is caused mainly by the coolant flow rate decrease or the pressure drop. It is the recirculation

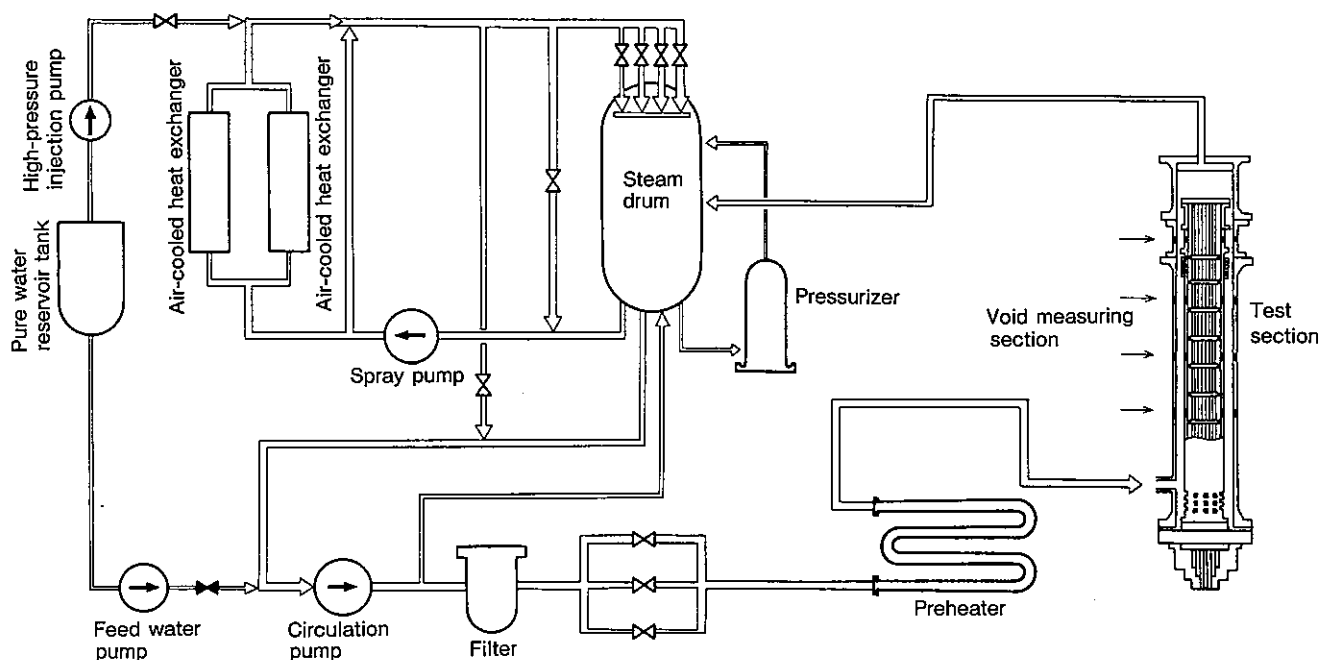


Fig. 2 · 2 System diagram of the BWR fuel assembly void measurement test facility

pump seizure that rapidly reduces the coolant flow rate, in which the coolant flow in the recirculation loop in trouble instantly reduces to almost zero. Further the recirculation pump trip is a typical event of abnormal transient in which the coolant flow rate decreases. The typical phenomenon causing the pressure to drop is the pressure controller failure (demand for increase).

The following evaluation is performed based on the data obtained from the void measurement tests under steady and transient conditions. **Figure 2 • 1** shows the test evaluation procedure.

- 1) Estimated void fraction values decided by the void correlations are compared with the measured values obtained from the tests to prove the reliability of the void correlations.

- 2) Void data obtained from the tests are rearranged in void tables according to each test parameter, in order to establish the void data base.

- 3) Regarding the void behaviors under transient conditions, the test data are compared with the estimated values by analysis to prove the reliability of the transient analysis and evaluation method and establish the data base to improve the codes in future.

- 4) The estimated values of void fraction distributions obtained by the subchannel analysis codes are compared with the test data to confirm the reliability of the analysis methods.

[3] Outline of Test Facility

The BWR fuel assembly maximum thermal loading test facility, which can simulate the same high-temperature and high-pressure conditions as in the actual reactor, is partially remodelled to use as the BWR fuel assembly void measurement test facility. The main remodelled section is the test section and the void fraction measuring equipment.

Figure 2 • 2 shows the system diagram of the BWR fuel assembly void measurement test facility.

The test facility consists of the test section that can simulate the same high-temperature and high-pressure conditions as in the actual reactor, the circulation system that supplies high-temperature and high-pressure water to the test section, the cooling system that removes the heat generated from the test section, etc.

The test section is composed of the pressure vessel, channel box, electrodes, etc. and loaded with an indirect type electrically heated full-size 8×8 mock-up fuel assembly (test assembly) that can simulate the same heated conditions as in the actual reactor. **Figure 2 • 3** shows the cross-sectional view of the test section.

An X-ray CT scanner is used to measure the detailed void fraction distribution in the flow area at the

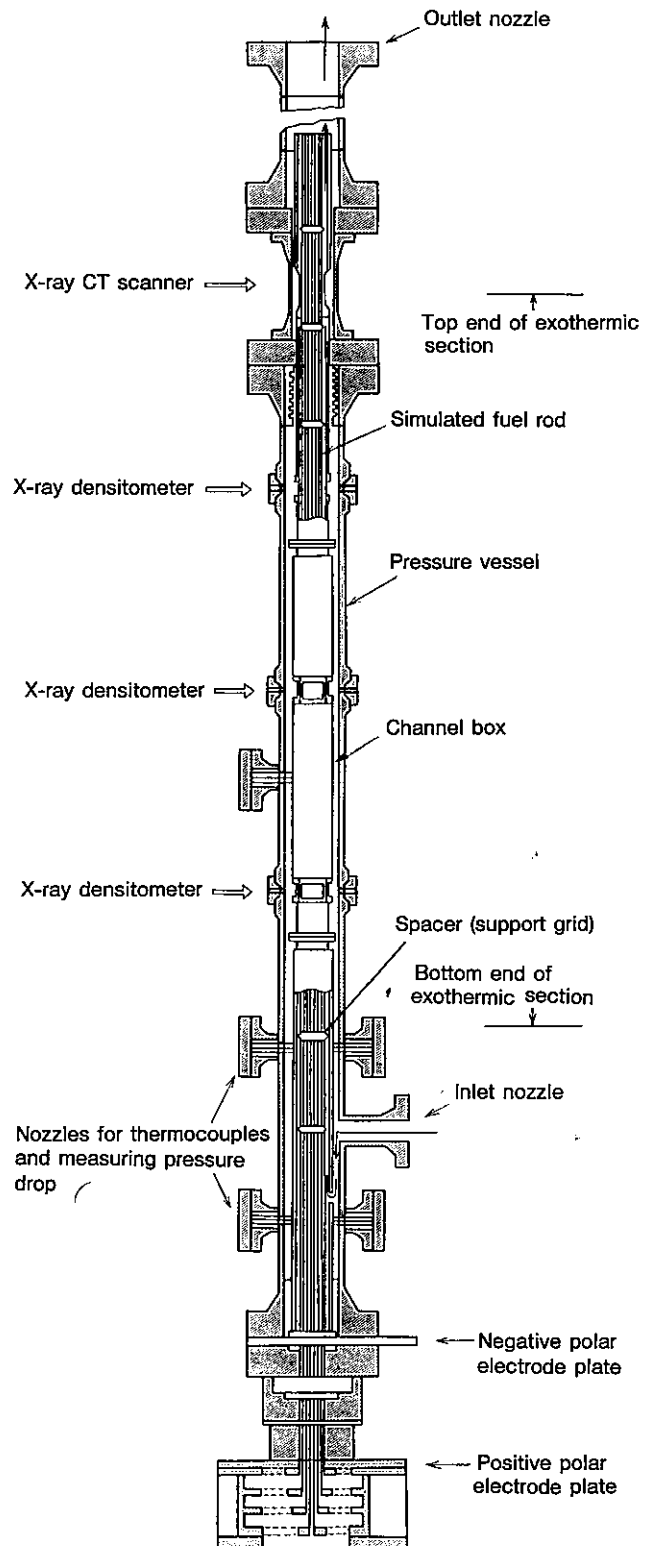


Fig. 2 • 3 Cross-sectional view showing the test section of the BWR fuel assembly void measurement test facility

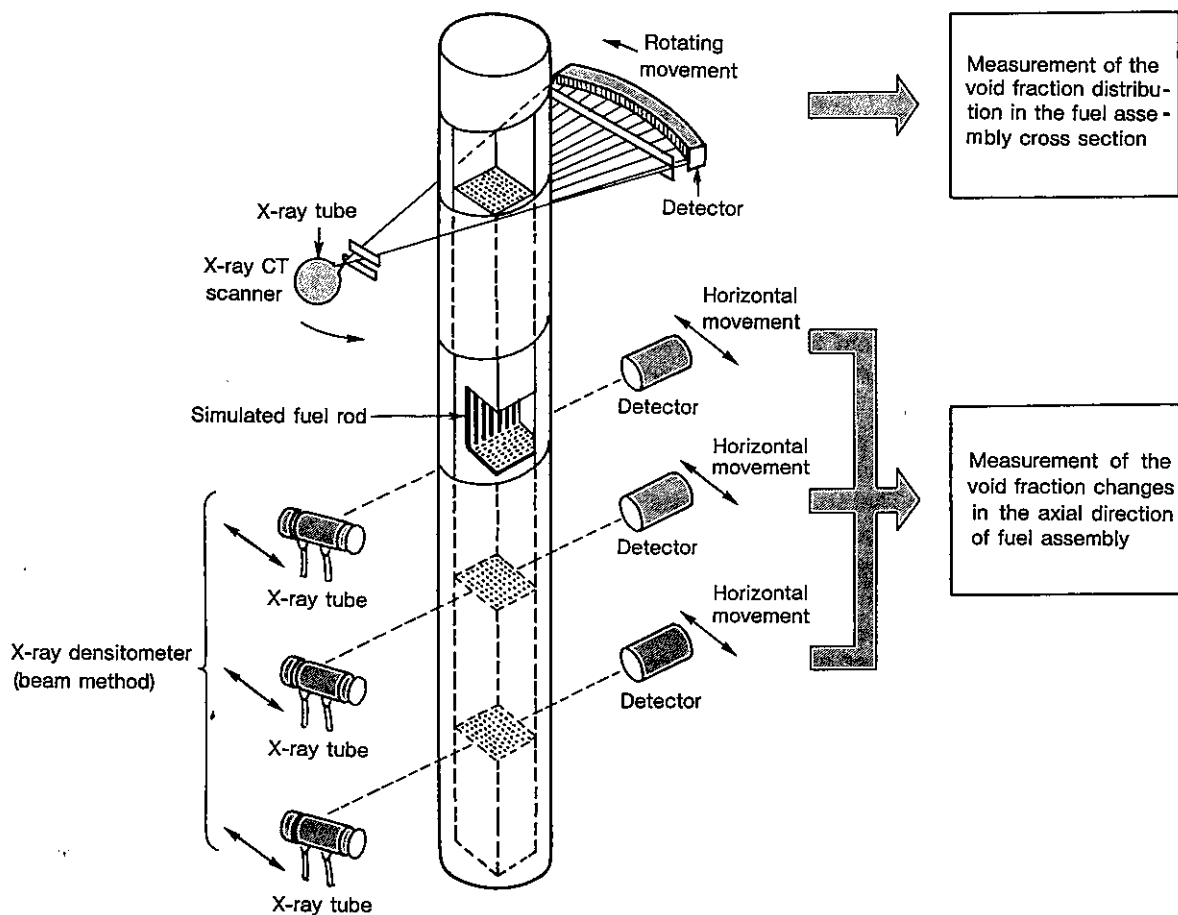


Fig. 2・4 Concept of the void measuring equipment of BWR fuel assembly void measurement test facility

exothermic end of the test assembly. The X-ray CT scanner irradiates X-rays to the object from various directions around it, measures the X-rays having penetrated through the object and draws up its sectional view through calculations. This equipment is now widely used in the medical treatment field, etc. Further three X-ray densitometers are installed at regular intervals in the axial direction to measure the axial void fraction distribution under steady conditions and the changes with the passage of time in that under transient conditions. **Figure 2・4** shows the concept of the void measuring equipment. **Figure 2・5** shows examples of the axial void fraction distribution under BWR rated operating conditions.

Further the thermocouples for measuring the fuel rod surface temperatures, the instruments for measuring the pressure, coolant flow rate, etc. are installed.

In the circulation system, the circulation water leaving the circulation pump passes through the filter, is heated by the preheater up to the set entrance temperature and then enters the lower end of the test section with its flow rate kept as prescribed by one of the

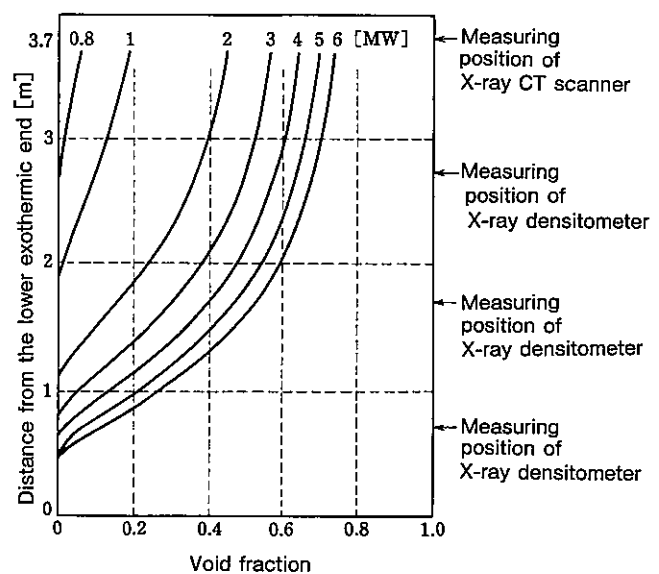


Fig. 2・5 Examples of the axial void fraction distribution under BWR rated operating conditions

control valves in the three flow control lines. The circulation water is heated and boiled while rising through the test section, enters the steam drum in the form of a steam/water mixed two-phase flow and is separated into steam and water by the separator built in the steam drum. Part of the water separated within the steam drum circulates through the cooling system and the remaining flows into the circulation pump again. By the way, the pressurizer installed in parallel with the steam drum is used as a steam generator to control the system pressure in the region where the pressure is uncontrollable by the spray valve when the heat power to the test section is insufficient.

In the cooling system, part of the water leaving the spray pump is bypassed and the remaining is cooled by the air-cooled heat exchanger. The cooled water and the water which bypassed the air-cooled heat exchanger join in the mixer of the cooling system and the spray water with a desired temperature is made. Part of the water circulating through the cooling system flows into the circulation system via its mixer and is used for securing the subcool at the inlet of the circulation pump. The water used as the spray of the steam drum is adjusted by the four spray flow control line valves, and the remaining passes through the spray bypass and returns to the spray pump via the mixer of the cooling system.

2. Void Measurement Test in PWR Fuel Assembly

[1] Outline of the Test

This test is intended to measure the void fraction distribution in the fuel assembly using a thermal hydraulic test facility that can simulate the same high-temperature and high-pressure conditions as in the actual reactor and prove the reliability of the PWR nuclear thermal hydraulic design method.

For this test, a part of the PWR fuel assembly maximum thermal loading test facility at Takasago Engineering Laboratory is remodelled so as to be suitable for the void fraction measurement. The test is made on electrically heated 5×5 mock-up fuel assemblies (test assemblies) that can simulate the same heated conditions as 17×17 type fuel assemblies in the actual reactor.

[2] Contents of the Test

In the PWR core, which uses the pressurized water, there is little void during normal operation, but there would be remarkable void in high power fuel assemblies at the critical operation state like abnormal transients and accidents. In this test, therefore, abnormal transients and accidents will be studied and the following tests are planned.

(a) Void measurement test under steady conditions

Among abnormal transients and accidents main steam line break event which is regarded as quasi-steady condition because of the relatively small change in the void fraction with the passage of time, etc. are simulated. The test is carried out under a wide range of test conditions and enough measurement time to obtain more accurate void data. After the power is raised by step and then kept at a fixed level when the pressure, flow rate and entrance temperature are kept steady, the fuel assembly void distribution is measured. The test conditions are as follows.

1) **Pressure : $60\text{--}170$ [kg/cm²]** The lower limit is set considering main steam line break conditions, and the upper limit is within the set pressure of the pressurizer safety valve.

2) **Coolant flow rate : $5\text{--}44$ [t/h]** The lower limit is set considering the loss of external power source, and the upper limit is set above the flow rate in pump

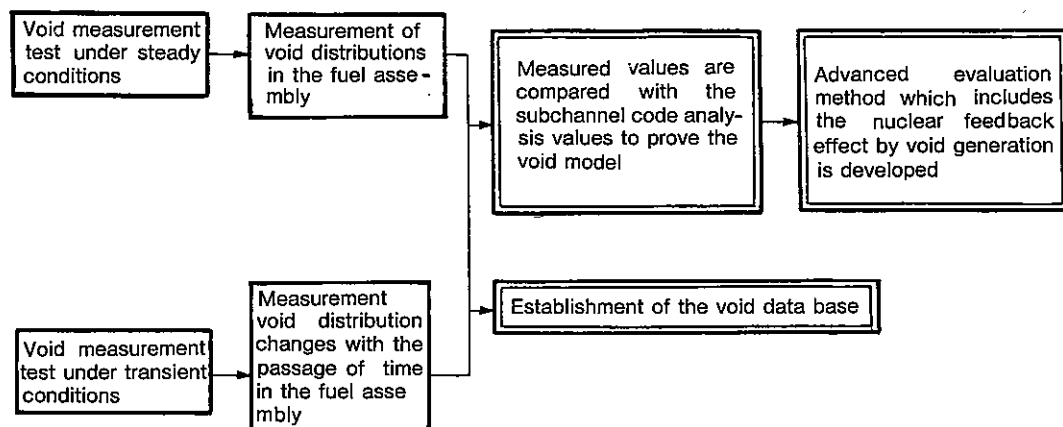


Fig. 2-6 Evaluation procedure for PWR fuel assembly void measurement test

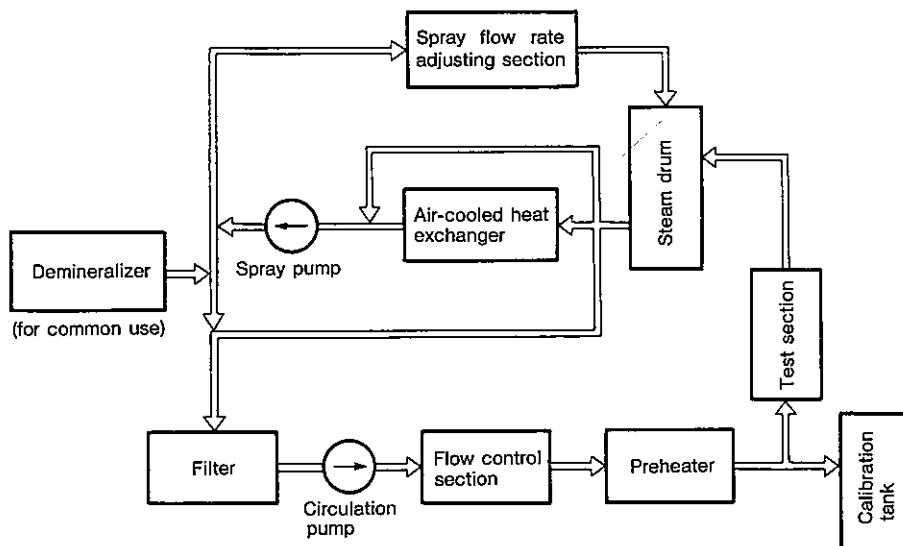


Fig. 2-7 System diagram of PWR fuel assembly void measurement test facility

overspeed.

3) **Entrance temperature : 200-325 [°C]** The lower limit is set considering main steam line break event, and the upper limit envelops the maximum temperature of the limit value for DNB (Departure from Nucleate Boiling) on the critical operation curves.

(b) Void measurement test under transient conditions

The power increase, flow rate reduction, depressurization and coolant temperature increase are considered as factors to increase void generation. In this test, a typical phenomena are selected from abnormal transients and accidents in respect to each of the above factors, and simulated to measure the change of void fraction distribution with the passage of time in the fuel assembly. The typical phenomena are as follows.

- (1) Event of power increase
 - * Fast control rod withdrawal
- (2) Event of flow rate reduction
 - * Loss of coolant flow rate
 - * Locked rotor of reactor coolant pump
- (3) Event of depressurization
 - * Abnormal depressurization in the reactor coolant system
- (4) Event of increase of coolant temperature
 - * Main feed water line break

The following evaluation, etc. are carried out using the data obtained from the above tests. **Figure 2-6** shows the evaluation procedure for void measurement test.

- 1) The reliability of the void correlation (void model) is proved in the fuel assembly to improve the estimating precision of the subchannel code.
- 2) The void correlation (void model) is verified in the fuel assembly to develop advanced evaluation method which includes the nuclear feedback effect by

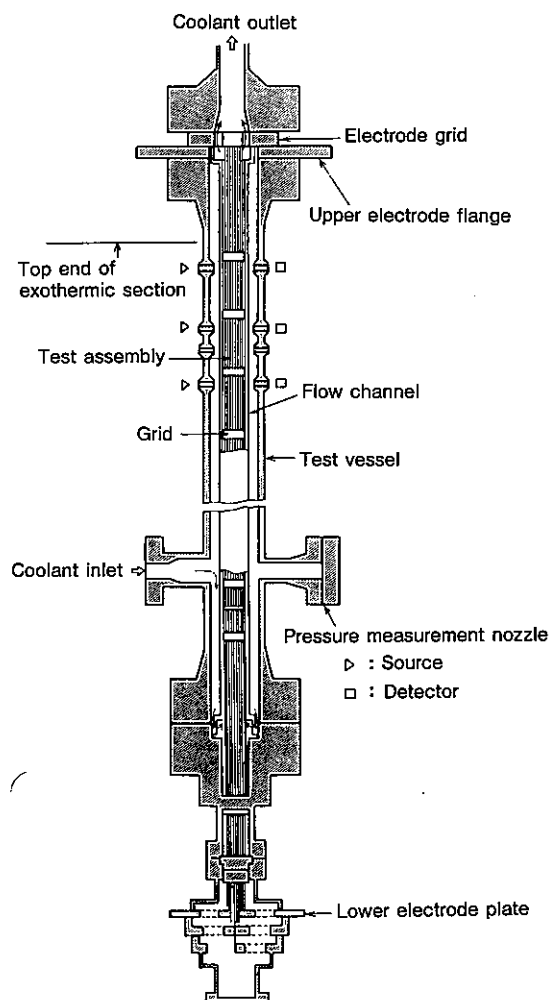


Fig. 2-8 Cross sectional view showing the test section of PWR fuel assembly void measurement test facility

void generation.

3) The data base on void phenomena for PWR core geometry is established.

[3] Outline of Test Facility

The PWR fuel assembly maximum thermal loading test facility, which can simulate the same high-

temperature and high-pressure conditions as in the actual reactor, is partially remodelled to use as the PWR fuel assembly void measurement test facility.

The main remodelled sections are the test section and the void measuring equipment.

Figure 2・7 shows the system diagram and **Figure 2・8** shows the cross sectional view of the test section.